A STABILIZER-FREE WEAK VIRTUAL ELEMENT METHOD FOR STOKES EQUATIONS WITH STRONGLY SYMMETRIC STRESS

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Abstract. We develop a stabilizer-free weak virtual element method for the incompressible Stokes equations on two-dimensional polygonal meshes using a strongly symmetric local viscous-stress reconstruction. The velocity combines an $H(\mathrm{div})$-conforming virtual field with single-valued tangential edge unknowns, and the pressure is discontinuous. The local stress is eliminated exactly, leaving a velocity–pressure formulation whose viscous bilinear form is determined solely by a weak symmetric strain. Stability follows from an intrinsic bubble–moment condition that supplies stable local stress liftings and a discrete Korn inequality. We establish the condition on shape-regular convex polygons for $k \geq 2$ using a centroid-based cone subdivision; a lowest-order local tensor enrichment provides the corresponding construction for $k = 1$. The discrete velocity is exactly divergence-free elementwise, and we derive stability and optimal-order velocity and pressure estimates under the stated hypotheses. Numerical experiments examine the convergence of the method on polygonal meshes.

Key words. stabilizer-free virtual element method, Stokes equations, exact divergence-free property, weak symmetric gradient, discrete Korn inequality, polygonal meshes, strongly symmetric local stress

1. Introduction.

Let $\Omega \subset \mathbb{R}^2$ be a polygonal domain. The incompressible Stokes equations may be expressed using the symmetric Cauchy stress $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\boldsymbol{I}$, where $\boldsymbol{\tau} = \varepsilon(\boldsymbol{u})$ is the viscous stress under the normalized viscosity convention adopted here. This formulation separates the symmetric viscous response from the pressure and provides a natural starting point for constructing locally symmetric numerical stresses. The central challenge on polygonal meshes is to combine such a reconstruction with exact incompressibility and a stable discretization without an extrinsic stabilization term.

Virtual element methods admit flexible polygonal discretizations while retaining computable polynomial projections [1]. $H(\mathrm{div})$-conforming virtual spaces [2] provide continuous normal velocity traces and polynomial divergence. The weak virtual element method of Chen and Wang [6] supplements such a velocity with tangential edge unknowns; its stability analysis uses an additional stabilization term. Stabilizer-free weak Galerkin and virtual element constructions [9, 7] show how suitably designed reconstruction spaces can control degrees of freedom not captured by standard polynomial projections.

We construct a local weak symmetric strain in a broken polynomial symmetric-tensor space on the cones joining a polygonal element's interior point to its edges. Defining the discrete viscous stress as this weak strain gives an equivalent local three-field formulation and a reduced velocity–pressure method. The reconstruction alone defines the viscous energy. Its stability is characterized by an intrinsic bubble–moment condition, which gives unisolvence, a stable stress lifting, and discrete Korn control. For convex polygons the condition holds for $k \geq 2$ when the cone apex is the centroid.

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The lowest-order space is augmented by a local normal–tangential bubble, allowing the same mechanism to apply for $k = 1$.

The resulting method has an exactly divergence-free discrete velocity, and its stability and optimal-order error estimates follow from the local reconstruction and the standard velocity–pressure inf–sup condition. Our analysis is restricted to two-dimensional polygonal meshes. The three-dimensional case requires separate treatment of polyhedral geometry and the associated local tensor spaces.

The remainder of the paper is organized as follows. Section 2 defines the polygonal meshes and velocity spaces. Section 3 constructs and analyzes the local symmetric-tensor reconstruction. Section 4 gives the scheme and its error estimates. Numerical results are discussed in Section 5.

2. VEM spaces. Let $\Omega \subset \mathbb{R}^2$ be a bounded and simply connected polygonal domain. Given an external force field $\boldsymbol{f} \in \boldsymbol{L}^2(\Omega)$, we consider the steady-state incompressible Stokes problem:

$$\begin{cases} -\operatorname{div}(\varepsilon(\boldsymbol{u})) + \nabla p = \boldsymbol{f} & \text{in } \Omega, \\ \operatorname{div} \boldsymbol{u} = 0 & \text{in } \Omega, \\ \boldsymbol{u} = \boldsymbol{0} & \text{on } \partial\Omega. \end{cases} \tag{2.1}$$

Here $\varepsilon(\boldsymbol{u}) := (\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}})/2$ is the symmetric strain, and the pressure is normalized by $\int_\Omega p \, \mathrm{d}x = 0$. We use $(\cdot,\cdot)_K$ and $\langle \cdot,\cdot \rangle_{\partial K}$ for the volume and boundary L^2 inner products, and $\| \cdot \|_K$ and $\| \cdot \|_{\partial K}$ for their norms.

2.1. Primal and dual meshes. Let $\mathcal{K}_h$ be a conforming mesh of convex polygons in Ω. Each element K is star-shaped with respect to a ball of radius at least ρh_K, where $h_K = \operatorname{diam} K$ and $\rho > 0$ is independent of $h := \max_K h_K$. We choose $\boldsymbol{x}_K$ to be the centroid of K. We assume that this point is uniformly separated from every edge in the shape-regular mesh family. We assume a uniformly bounded number of faces per element. For each edge $F \subset \partial K$, define

$$T_F = \operatorname{conv}(\boldsymbol{x}_K, F).$$

We refer to T_F as a cone, which is a triangle in two dimensions. These cones form a subdivision $\mathcal{T}_h$ of the polygonal mesh. We assume that these cones admit a conforming, uniformly shape-regular simplicial refinement with simplex diameters comparable to h_K. In particular, $h_F := \operatorname{diam} F$ is comparable to h_K. These assumptions are used in the trace, inverse, and scaling estimates below.

Let $\mathcal{F}_h$ and $\mathcal{F}_h^T$ be the face sets of $\mathcal{K}_h$ and $\mathcal{T}_h$, respectively. Their interior faces are denoted by $\mathring{\mathcal{F}}_h^K$ and $\mathring{\mathcal{F}}_h^T$. Let $\mathcal{T}_h(K)$ be the set of cones in K.

Let $F \in \partial K$ be an edge with unit normal $\boldsymbol{n}_F$ and orthonormal tangent vectors $\{\boldsymbol{t}_i^F\}_{i=1}^{d-1}$. For any $\boldsymbol{v} \in \mathbb{R}^d$, the tangential part on F is

$$\Pi_F \boldsymbol{v} = (\boldsymbol{I} - \boldsymbol{n}_F \otimes \boldsymbol{n}_F)\boldsymbol{v} = \sum_{i=1}^{d-1} (\boldsymbol{v} \cdot \boldsymbol{t}_i^F)\boldsymbol{t}_i^F.$$

With $\operatorname{sym} A = (A + A^{\mathsf{T}})/2$ and $\operatorname{skw} A = (A - A^{\mathsf{T}})/2$, define

$$\begin{aligned} \mathscr{T}^F &:= \operatorname{span}\{\boldsymbol{t}_1^F, \dots, \boldsymbol{t}_{d-1}^F\}, \\ \mathscr{T}^F(\mathbb{S}) &:= \operatorname{span}\{\operatorname{sym}(\boldsymbol{t}_i^F \otimes \boldsymbol{t}_j^F) \quad \text{for } 1 \le i \le j \le d-1\}, \\ \mathscr{N}^F(\mathbb{S}) &:= \operatorname{sym}(\mathscr{T}^F \otimes \boldsymbol{n}_F) \oplus \operatorname{span}\{\boldsymbol{n}_F \otimes \boldsymbol{n}_F\}, \\ \mathscr{T}^F(\mathbb{K}) &:= \operatorname{span}\{\operatorname{skw}(\boldsymbol{t}_i^F \otimes \boldsymbol{t}_j^F) \quad \text{for } 1 \le i < j \le d-1\}. \end{aligned}$$

These subspaces give the following $\boldsymbol{t}$-$\boldsymbol{n}$ decompositions

$$\mathbb{S} = \mathscr{T}^F(\mathbb{S}) \oplus \mathscr{N}^F(\mathbb{S}), \tag{2.2}$$

$$\mathbb{K} = \mathscr{T}^F(\mathbb{K}) \oplus \mathscr{N}^F(\mathbb{K}), \quad \mathscr{N}^F(\mathbb{K}) = \mathrm{skw}(\mathscr{T}^F \otimes \boldsymbol{n}_F). \tag{2.3}$$

Fix an integer $k \geq 1$. For $K \in \mathcal{K}_h$, let $\mathbb{P}_k(K)$ be the space of polynomials on K of degree at most k. Let $\mathbb{P}_k(K;\mathbb{X}) = \mathbb{P}_k(K) \otimes \mathbb{X}$, where $\mathbb{X}$ is $\mathbb{R}^d$ for vectors, $\mathbb{S}$ for symmetric matrices, and $\mathbb{K}$ for skew-symmetric matrices.

We set $\mathbb{P}_j = \{0\}$ for $j < 0$, and $\mathbf{RM}(K) = \{\boldsymbol{a} + \boldsymbol{B}\boldsymbol{x} : \boldsymbol{a} \in \mathbb{R}^d,\ \boldsymbol{B} \in \mathbb{K}\}$.

2.2. VEM spaces. The local velocity space and its degrees of freedom are the $H(\mathrm{div})$ virtual elements of [2]. On each element $K \in \mathcal{K}_h$, we introduce a local space

$$\boldsymbol{V}_K = \left\{ \begin{array}{l} \boldsymbol{v} \in \boldsymbol{H}(\mathrm{div}, K) \cap \boldsymbol{H}(\mathrm{curl}, K) : \boldsymbol{v} \cdot \boldsymbol{n} \in \mathbb{P}_k(F)\ \forall F \subset \partial K, \\ \mathrm{div}\, \boldsymbol{v} \in \mathbb{P}_{k-1}(K),\ \mathrm{curl}\, \boldsymbol{v} \in \mathrm{curl}\, \mathbb{P}_k(K;\mathbb{R}^d) \end{array} \right\},$$

The global virtual element space is

$$\boldsymbol{V}_h^{\mathrm{div}} := \{\boldsymbol{v} \in \boldsymbol{H}(\mathrm{div}, \Omega), \boldsymbol{v}|_K \in \boldsymbol{V}_K\ \forall K \in \mathcal{K}_h, \boldsymbol{v} \cdot \boldsymbol{n} = 0 \text{ on } \partial\Omega\}.$$

The degrees of freedom (DoFs) for $\boldsymbol{V}_K$ are

$$\text{Type I} \qquad \int_F \boldsymbol{v} \cdot \boldsymbol{n} q_k \,\mathrm{d}s \qquad \forall q_k \in \mathbb{P}_k(F), F \subset \partial K,$$

$$\text{Type II} \qquad \int_K \boldsymbol{v} \cdot \boldsymbol{q}_{k-2} \,\mathrm{d}\boldsymbol{x} \qquad \forall \boldsymbol{q}_{k-2} \in \mathbf{G}_{k-2}(K),$$

$$\text{Type III} \qquad \int_K \boldsymbol{v} \cdot \boldsymbol{q}_k \,\mathrm{d}\boldsymbol{x} \qquad \forall \boldsymbol{q}_k \in \mathbf{G}_k(K)^{\perp},$$

where $\mathbf{G}_k(K) = \nabla \mathbb{P}_{k+1}(K)$. Here $\mathbf{G}_k(K)^{\perp}$ is its $L^2(K)$-orthogonal complement in $\mathbb{P}_k(K;\mathbb{R}^d)$. Moments are taken against a basis of each listed space. These DoFs are unisolvent [2].

On each face $F \in \mathcal{F}_h$, we introduce a space

$$\boldsymbol{V}_F := \mathbb{P}_{k-1}(F; \mathscr{T}^F) + \mathbf{RM}(F),$$

where $\mathbf{RM}(F)$ denotes the space of rigid motions, i.e., $\mathbf{RM}(F) := \Pi_F \mathbf{RM}(K) = \mathbb{P}_0(F; \mathscr{T}^F) \oplus \mathbb{P}_0(F; \mathscr{T}^F(\mathbb{K}))\boldsymbol{x}$. The DoFs for functions in $\boldsymbol{V}_F$ are

$$\int_F \boldsymbol{v} \cdot \boldsymbol{q} \,\mathrm{d}s \quad \forall \boldsymbol{q} \in \mathbb{P}_{k-1}(F; \mathscr{T}^F) + \mathbf{RM}(F).$$

The tangential face space is

$$\boldsymbol{V}_h^t := \{\boldsymbol{v}^t, \boldsymbol{v}^t|_F \in \boldsymbol{V}_F, F \in \mathcal{F}_h, \boldsymbol{v}^t = \boldsymbol{0} \text{ on } \partial\Omega\}.$$

The weak virtual element space is

$$\boldsymbol{V}_h := \left\{ \{\boldsymbol{v}_h^{\mathrm{div}}, \boldsymbol{v}_h^t\}, \boldsymbol{v}_h^{\mathrm{div}} \in \boldsymbol{V}_h^{\mathrm{div}}, \boldsymbol{v}_h^t \in \boldsymbol{V}_h^t \right\}.$$

2.3. Projection operator. For $\boldsymbol{v} \in \boldsymbol{L}^2(K)$, let $\boldsymbol{\Pi}_K^o \boldsymbol{v} \in \mathbb{P}_k(K;\mathbb{R}^d)$ satisfy

$$\int_K \boldsymbol{\Pi}_K^o \boldsymbol{v} \cdot \boldsymbol{q}\,\mathrm{d}x = \int_K \boldsymbol{v} \cdot \boldsymbol{q}\,\mathrm{d}x \quad \forall \boldsymbol{q} \in \mathbb{P}_k(K;\mathbb{R}^d).$$

For $\boldsymbol{v} \in \boldsymbol{V}_K$, Type III DoFs give the moments in $\mathbf{G}_k(K)^\perp$. The remaining moments follow by integration by parts: for $p_{k+1} \in \mathbb{P}_{k+1}(K)$,

$$\int_K \boldsymbol{v} \cdot \nabla p_{k+1}\,\mathrm{d}\boldsymbol{x} = -\int_K \operatorname{div} \boldsymbol{v}\, p_{k+1}\,\mathrm{d}\boldsymbol{x} + \int_{\partial K} \boldsymbol{v} \cdot \boldsymbol{n}\, p_{k+1}\,\mathrm{d}s.$$

The normal trace is determined by Type I DoFs. The divergence is determined by Type I and II DoFs through

$$(\operatorname{div} \boldsymbol{v}, q)_K = -(\boldsymbol{v}, \nabla q)_K + \langle \boldsymbol{v} \cdot \boldsymbol{n}, q\rangle_{\partial K}, \qquad q \in \mathbb{P}_{k-1}(K).$$

Thus every term above is computable [2]. We write $(\boldsymbol{\Pi}_h^o \boldsymbol{v})|_K = \boldsymbol{\Pi}_K^o(\boldsymbol{v}|_K)$.

3. Weak strain.

3.1. Weak strain space. Let $T \in \mathcal{T}_h(K)$ have base face F. The local weak strain space is

$$\begin{aligned}\Sigma_k(T) = {} & \mathbb{P}_k(T)\boldsymbol{n}_F \otimes \boldsymbol{n}_F \\ & \oplus \operatorname{sym}\big(\boldsymbol{n}_F \otimes [\mathbb{P}_{k-1}(T;\mathscr{T}^F) + \mathbf{RM}(F)]\big) \\ & \oplus \mathbb{P}_{k-1}(T;\mathscr{T}^F(\mathbb{S})).\end{aligned} \tag{3.1}$$

Face polynomials, including $\mathbf{RM}(F)$, are extended constantly along lines normal to F. For $k = 1$, the middle block is $\operatorname{sym}(\boldsymbol{n}_F \otimes \mathbf{RM}(F))$; for $k > 1$, it is $\operatorname{sym}(\boldsymbol{n}_F \otimes \mathbb{P}_{k-1}(T;\mathscr{T}^F))$. Set

$$V_k(F) := \mathbb{P}_k(F)\boldsymbol{n}_F \oplus \boldsymbol{V}_F.$$

The normal trace $\operatorname{tr}_F \boldsymbol{\sigma} := \boldsymbol{\sigma}\boldsymbol{n}_F|_F$ satisfies $\operatorname{tr}_F \Sigma_k(T) = V_k(F)$. In particular, $\mathbb{P}_{k-1}(K;\mathbb{S})$ is contained in the piecewise strain space on K.

We define the local symmetric tensor space on K as the discontinuous polynomial space

$$\Sigma_h^{-1}(K) = \textstyle\bigoplus_{i=1}^n \chi_{T_i}\Sigma_k(T_i) = \{\boldsymbol{\tau} \in L^2(K;\mathbb{S}) : \boldsymbol{\tau}|_T \in \Sigma_k(T) \text{ for } T \in \mathcal{T}_h(K)\}, \tag{3.2}$$

where χ_{T_i} is the characteristic function of T_i. We define the bubble space

$$\begin{aligned}\mathbb{B}_k(\operatorname{div}, K;\mathbb{S}) = {} & \{\boldsymbol{\tau} \in \Sigma_h^{-1}(K) : \boldsymbol{\tau}\boldsymbol{n}|_{\partial K} = \mathbf{0}\} \\ = {} & \textstyle\bigoplus_{i=1}^n \lambda_{\boldsymbol{x}_K}\chi_{T_i}(\mathbb{P}_{k-1}(T_i)\boldsymbol{n}_i \otimes \boldsymbol{n}_i) \\ & \oplus \textstyle\bigoplus_{i=1}^n \lambda_{\boldsymbol{x}_K}\chi_{T_i} \operatorname{sym}(\mathbb{P}_{k-2}(T_i;\mathscr{T}^{F_i}) \otimes \boldsymbol{n}_i) \\ & \oplus \textstyle\bigoplus_{i=1}^n \chi_{T_i}\mathbb{P}_{k-1}(T_i;\mathscr{T}^{F_i}(\mathbb{S})).\end{aligned} \tag{3.3}$$

Here F_i is the base face of T_i, $\boldsymbol{n}_i$ is its outward normal, and $\lambda_{\boldsymbol{x}_K}$ is the affine function on T_i that equals one at $\boldsymbol{x}_K$ and zero on F_i. The bubble space imposes no continuity across the internal faces of K.

Assumption 3.1 (Intrinsic bubble–moment stability). There is a constant $\beta_* > 0$, independent of h and K, such that

$$\inf_{\mathbf{0}\neq\boldsymbol{q}\in\mathbb{P}_{k-1}(K;\mathbb{S})} \sup_{\mathbf{0}\neq\boldsymbol{\tau}\in\mathbb{B}_k(\operatorname{div},K;\mathbb{S})} \frac{(\boldsymbol{\tau},\boldsymbol{q})_K}{\|\boldsymbol{\tau}\|_K\,\|\boldsymbol{q}\|_K} \geq \beta_*. \tag{3.4}$$

The constant may depend on k and the mesh-regularity parameters, but not on h or on the particular element K.

Assumption 3.1 is intrinsic to the local reconstruction: it requires the zero-traction bubble space to detect every polynomial symmetric tensor needed in the volume moments. It does not select preferred faces or tensor directions. Define the moment operator

$$\mathcal{M}_K : \mathbb{B}_k(\mathrm{div}, K; \mathbb{S}) \longrightarrow \mathbb{P}_{k-1}(K; \mathbb{S})^*, \qquad (\mathcal{M}_K \boldsymbol{\tau})(\boldsymbol{q}) = (\boldsymbol{\tau}, \boldsymbol{q})_K.$$

Then (3.4) is equivalent, after scaling to a reference-size element, to surjectivity of $\mathcal{M}_K$ with a uniformly bounded right inverse. Set

$$\mathcal{Z}_K := \ker \mathcal{M}_K, \qquad \mathcal{E}_K := \mathcal{Z}_K^{\perp} \cap \mathbb{B}_k(\mathrm{div}, K; \mathbb{S}),$$

where orthogonality is in $L^2(K; \mathbb{S})$. Consequently

$$\mathbb{B}_k(\mathrm{div}, K; \mathbb{S}) = \mathcal{Z}_K \oplus \mathcal{E}_K, \qquad \dim \mathcal{E}_K = \dim \mathbb{P}_{k-1}(K; \mathbb{S}),$$

and the restriction of $\mathcal{M}_K$ to $\mathcal{E}_K$ is an isomorphism with uniformly bounded inverse.

Remark 3.2 (Relation with the face-spanning condition). A sufficient, but generally stronger, way to verify Assumption 3.1 is to select $d(d+1)/2$ uniformly well-conditioned tensors from the face-tangential spaces $\mathscr{T}^F(\mathbb{S})$ that span $\mathbb{S}$. The corresponding functions $\chi_T p\boldsymbol{b}$, $p \in \mathbb{P}_{k-1}(T)$, belong to the last summand of the bubble space and give a uniformly nondegenerate pairing with $\mathbb{P}_{k-1}(K; \mathbb{S})$. This face-spanning criterion, used as a geometric hypothesis in an earlier formulation of the analysis, is made precise in Proposition 3.3. It gives a sufficient construction of a stable right inverse of $\mathcal{M}_K$, but is not required by the present intrinsic bubble-moment condition.

Proposition 3.3 (A sufficient face-spanning criterion). *Suppose that, for every $K \in \mathcal{K}_h$, there are $d(d+1)/2$ unit tensors $\{\boldsymbol{b}_\ell\}_{\ell=1}^{d(d+1)/2}$, selected from the face-tangential spaces $\mathscr{T}^F(\mathbb{S})$, which form a uniformly well-conditioned basis of $\mathbb{S}$. Then Assumption 3.1 holds.*

Proof. Let $\{\boldsymbol{b}_\ell^*\}$ be the dual tensor basis and write

$$\boldsymbol{q} = \sum_{\ell=1}^{d(d+1)/2} q_\ell \boldsymbol{b}_\ell^*, \qquad q_\ell \in \mathbb{P}_{k-1}(K).$$

Choose, for each ℓ, a cone T_ℓ whose base face has $\boldsymbol{b}_\ell \in \mathscr{T}^{F_\ell}(\mathbb{S})$, and set

$$\boldsymbol{\tau} = \sum_{\ell=1}^{d(d+1)/2} \chi_{T_\ell} q_\ell \boldsymbol{b}_\ell.$$

Since the last summand in the definition of $\mathbb{B}_k(\mathrm{div}, K; \mathbb{S})$ contains $\chi_{T_\ell}\mathbb{P}_{k-1}(T_\ell; \mathscr{T}^{F_\ell}(\mathbb{S}))$, we have $\boldsymbol{\tau} \in \mathbb{B}_k(\mathrm{div}, K; \mathbb{S})$. By duality,

$$(\boldsymbol{\tau}, \boldsymbol{q})_K = \sum_\ell \|q_\ell\|_{T_\ell}^2.$$

For fixed k, polynomial norm equivalence on a cone of diameter comparable with h_K, the mesh-regularity assumptions, and the uniform conditioning of $\{\boldsymbol{b}_\ell\}$ give

$$\|\boldsymbol{q}\|_K^2 \lesssim \sum_\ell \|q_\ell\|_{T_\ell}^2, \qquad \|\boldsymbol{\tau}\|_K^2 \lesssim \sum_\ell \|q_\ell\|_{T_\ell}^2.$$

Consequently

$$\sup_{\mathbf{0}\neq\boldsymbol{\tau}\in\mathbb{B}_k(\mathrm{div},K;\mathbb{S})} \frac{(\boldsymbol{\tau},\boldsymbol{q})_K}{\|\boldsymbol{\tau}\|_K} \gtrsim \|\boldsymbol{q}\|_K,$$

with a constant independent of h and K. ∎

LEMMA 3.4 (Two nonorthogonal face directions). *Let $d = 2$ and let K be a polygon. Suppose that K has two faces F and G whose tangent directions are neither parallel nor orthogonal. Then, for any $k \geq 1$, the moment operator*

$$\mathcal{M}_K : \mathbb{B}_k(\mathrm{div}, K; \mathbb{S}) \longrightarrow \mathbb{P}_{k-1}(K; \mathbb{S})^*$$

is surjective. Moreover, on a shape-regular family for which the two selected directions remain uniformly separated from the parallel and orthogonal configurations, the inf–sup constant in (3.4) *is uniform.*

Proof. Put $m = k-1$ and suppose that $\boldsymbol{q} \in \mathbb{P}_m(K;\mathbb{S})$ is orthogonal to the whole bubble space. Fix a face F, and let $(\boldsymbol{n}_F, \boldsymbol{t}_F)$ be its normal–tangent frame. The normal–normal bubble block gives

$$\int_{T_F} \lambda_{\boldsymbol{x}_K} (\boldsymbol{n}_F^{\mathsf{T}} \boldsymbol{q} \boldsymbol{n}_F) p = 0 \qquad \forall p \in \mathbb{P}_m(T_F).$$

Taking $p = \boldsymbol{n}_F^{\mathsf{T}}\boldsymbol{q}\boldsymbol{n}_F$ and using $\lambda_{\boldsymbol{x}_K} > 0$ in the interior of T_F yields $\boldsymbol{n}_F^{\mathsf{T}}\boldsymbol{q}\boldsymbol{n}_F = 0$ on T_F and hence, by polynomial continuation, on K. The tangential–tangential bubble block similarly gives $\boldsymbol{t}_F^{\mathsf{T}}\boldsymbol{q}\boldsymbol{t}_F = 0$ on K. Consequently

$$\boldsymbol{q} = s_F \ \mathrm{sym}(\boldsymbol{n}_F \otimes \boldsymbol{t}_F), \qquad s_F \in \mathbb{P}_m(K).$$

Applying the same argument to G gives

$$\boldsymbol{q} = s_G \ \mathrm{sym}(\boldsymbol{n}_G \otimes \boldsymbol{t}_G).$$

In two dimensions the one-dimensional tensor spaces $\mathrm{span}\{\mathrm{sym}(\boldsymbol{n}_F \otimes \boldsymbol{t}_F)\}$ and $\mathrm{span}\{\mathrm{sym}(\boldsymbol{n}_G \otimes \boldsymbol{t}_G)\}$ coincide if and only if the two tangent directions are parallel or orthogonal. By hypothesis their intersection is therefore $\{0\}$, and $\boldsymbol{q} = 0$. Hence the adjoint of $\mathcal{M}_K$ is injective and $\mathcal{M}_K$ is surjective.

For the uniform statement, scale K to unit diameter. The local moment matrix depends continuously on the normalized element geometry. The stated uniform angular separation and the mesh-regularity assumptions keep these geometries in a compact set on which the smallest singular value cannot vanish. Scaling back gives the asserted uniform bound. ∎

The next result is important because it verifies the intrinsic condition on a mesh family excluded by the face-spanning criterion. We first record a scalar polynomial lemma on the reference square.

LEMMA 3.5 (Weighted cone moments on a square). *Let $\widehat{K} = (-1,1)^2$ and let $\widehat{K}$ be divided by its center into the four triangles $\widehat{T}_R, \widehat{T}_L, \widehat{T}_T, \widehat{T}_B$ with bases on the right, left, top, and bottom sides. Let $\widehat{\lambda}_F$ be the affine function on $\widehat{T}_F$ which is one at the center and zero on its base. If $m \geq 1$ and $s \in \mathbb{P}_m(\widehat{K})$ satisfies*

$$\int_{\widehat{T}_F} \widehat{\lambda}_F s r \, \mathrm{d}\widehat{x} = 0 \qquad \forall r \in \mathbb{P}_{m-1}(\widehat{T}_F), \quad F = R, L, T, B, \tag{3.5}$$

then $s = 0$.

Proof. It is enough to use the right, left, and top triangles. On $\widehat{T}_R$ use the Duffy variables

$$x = \rho, \qquad y = \rho\eta, \qquad 0 < \rho < 1, \quad -1 < \eta < 1.$$

Then $\widehat{\lambda}_R = 1 - \rho$ and $\mathrm{d}x\,\mathrm{d}y = \rho\,\mathrm{d}\rho\,\mathrm{d}\eta$. The orthogonal complement of $\mathbb{P}_{m-1}(\widehat{T}_R)$ in $\mathbb{P}_m(\widehat{T}_R)$ for the weighted inner product $\int_{\widehat{T}_R} \widehat{\lambda}_R uv$ has the standard Duffy–Jacobi basis

$$\Phi_j^R(x,y) = x^j P_j(y/x) P_{m-j}^{(1,2j+1)}(2x-1), \qquad j = 0, \dots, m, \tag{3.6}$$

where P_j is the Legendre polynomial and $P_n^{(\alpha,\beta)}$ is the Jacobi polynomial. The left cone is the reflection of the right cone across $x = 0$. Consequently, if s satisfies the moment conditions on both cones, then both $s(x,y)$ and $s(-x,y)$ are orthogonal to $\mathbb{P}_{m-1}(\widehat{T}_R)$ with weight $1 - x$. Their difference is of the form $w(x,y) = s(x,y) - s(-x,y) = xr(x,y)$ with $r \in \mathbb{P}_{m-1}$. Since w satisfies the right-cone moment conditions, testing it with r yields

$$0 = \int_{\widehat{T}_R} (1-x)x\, r(x,y)^2 \,\mathrm{d}x\mathrm{d}y.$$

Both weights are strictly positive in the interior of $\widehat{T}_R$; hence $r = 0$ and s is even in x.

Expand $s = \sum_{j=0}^m c_j \Phi_j^R$ in the orthogonal basis (3.6). The polynomial $x^j P_j(y/x)$ is even in x and its coefficient of x^0 is $a_j y^j$, where $a_j \neq 0$ is the leading coefficient of P_j. For $j < m$ the radial factor $R_j(x) = P_{m-j}^{(1,2j+1)}(2x-1)$ has $R_j'(0) \neq 0$, by the Jacobi derivative identity

$$R_j'(0) = (m + j + 3) P_{m-j-1}^{(2,2j+2)}(-1) \neq 0.$$

Thus the coefficient of x^1 in s is exactly $\sum_{j=0}^{m-1} c_j a_j R_j'(0) y^j$. Because s is even in x, this polynomial is zero; linear independence of $1, y, \dots, y^{m-1}$ gives $c_0 = \cdots = c_{m-1} = 0$. For $j = m$ the radial Jacobi polynomial has degree zero, so $\Phi_m^R = x^m P_m(y/x)$. We have proved, without a coefficient-comparison shortcut, that

$$\left(\mathbb{P}_{m-1}^{\perp,R}\right) \cap \left(\mathbb{P}_{m-1}^{\perp,L}\right) = \mathrm{span}\{H_m\}, \qquad H_m(x,y) := x^m P_m(y/x). \tag{3.7}$$

Conversely, H_m belongs to both orthogonal complements: in Duffy variables it is $\rho^m P_m(\eta)$, and the angular Legendre orthogonality eliminates every test polynomial of total degree at most $m-1$; its reflection is itself.

Hence the right and left conditions in (3.5) imply $s = cH_m$. On the top triangle write $y = \rho$, $x = \rho\eta$. Since H_m is homogeneous,

$$H_m(\rho\eta, \rho) = \rho^m \widetilde{H}_m(\eta), \qquad \widetilde{H}_m(\eta) = \eta^m P_m(1/\eta),$$

where the right-hand side is understood as its polynomial continuation at $\eta = 0$. Testing the top condition with the homogeneous monomials $x^b y^{m-1-b}$, $b = 0, \dots, m-1$, and separating the positive radial factor gives

$$\int_{-1}^{1} \widetilde{H}_m(\eta)\eta^b \,\mathrm{d}\eta = 0, \qquad b = 0, \dots, m-1.$$

If m is odd, $\widetilde{H}_m \in \mathbb{P}_{m-1}(-1,1)$, so the last relations with the test function $\widetilde{H}_m$ imply $\widetilde{H}_m = 0$, a contradiction. If m is even, $\widetilde{H}_m$ has degree m, and the relations imply that it is a multiple of P_m. This is impossible: the zeros of P_m lie in $(-1,1)$, whereas the finite zeros of $\eta^m P_m(1/\eta)$ are the reciprocals of the nonzero zeros of P_m and therefore lie outside $[-1,1]$. Thus $c = 0$ and $s = 0$. ∎

THEOREM 3.6 (Verification on two-dimensional rectangles). *Let $d = 2$ and $k \geq 2$. On any shape-regular family of rectangular elements, choose the cone apex $\boldsymbol{x}_K$ to be the rectangle center. Then Assumption 3.1 holds for the bubble space defined above. The lower bound is uniform with respect to h and to the element, with a constant depending only on k and the shape-regularity bound.*

Proof. Put $m = k - 1 \geq 1$ and suppose that $\boldsymbol{q} \in \mathbb{P}_m(K;\mathbb{S})$ is orthogonal to $\mathbb{B}_k(\mathrm{div}, K;\mathbb{S})$. Rotate coordinates so that the rectangle sides are parallel to the coordinate axes. On a cone T_F with normal $\boldsymbol{n}_F$ and tangent $\boldsymbol{t}_F$, the normal–normal bubble block gives

$$\int_{T_F} \lambda_{\boldsymbol{x}_K} (\boldsymbol{n}_F^{\mathsf{T}} \boldsymbol{q} \boldsymbol{n}_F) p = 0 \qquad \forall p \in \mathbb{P}_m(T_F).$$

Taking $p = \boldsymbol{n}_F^{\mathsf{T}} \boldsymbol{q} \boldsymbol{n}_F$ shows that this component vanishes on T_F, hence, by polynomial continuation, on K. Likewise the tangential–tangential bubble block gives $\boldsymbol{t}_F^{\mathsf{T}} \boldsymbol{q} \boldsymbol{t}_F = 0$ on K. Therefore the diagonal components of $\boldsymbol{q}$ vanish and

$$\boldsymbol{q} = s\, \mathrm{sym}(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2) \qquad \text{for some } s \in \mathbb{P}_m(K).$$

The normal–tangential bubble block is present because $k \geq 2$ and yields, on each cone,

$$\int_{T_F} \lambda_{\boldsymbol{x}_K} s r = 0 \qquad \forall r \in \mathbb{P}_{m-1}(T_F).$$

Because the apex is the rectangle center, an affine scaling of the rectangle to $\widehat{K} = (-1,1)^2$ maps all four cones to the centered square cones. These are therefore exactly the conditions of Lemma 3.5; hence $s = 0$ and $\boldsymbol{q} = 0$. Thus the adjoint of the moment operator $\mathcal{M}_K$ is injective, equivalently $\mathcal{M}_K$ is surjective. For fixed k, the resulting smallest singular value on a unit-diameter rectangle depends continuously on its aspect ratio. Shape regularity keeps the aspect ratio in a compact subset of $(0,\infty)$, and the preceding injectivity shows that the singular value never vanishes there. Scaling back to K proves the uniform inf–sup bound (3.4). ∎

PROPOSITION 3.7 (Arbitrary cone apex on a rectangle for $k = 2$). *Let $K \subset \mathbb{R}^2$ be a rectangle and let a be any interior cone apex satisfying the uniform star-cone shape-regularity assumption. For $k = 2$ the local bubble-moment condition (3.4) holds, with a constant uniform on a shape-regular family of rectangles and admissible apices.*

Proof. Set $m = 1$. As in Theorem 3.6, normal–normal and tangential–tangential bubbles force any polynomial symmetric tensor in the kernel of the adjoint moment map to be $q = s\,\mathrm{sym}(e_1 \otimes e_2)$, where $s \in \mathbb{P}_1(K)$. The normal–tangential bubble block on each cone T_F includes $\lambda_a\, \mathrm{sym}(n_F \otimes t_F)$; hence $\int_{T_F} \lambda_a s = 0$ on all four cones. Write the cone vertices as a, b_F, c_F and let $m_F = (b_F + c_F)/2$ be the midpoint of its base. Barycentric integration on a triangle gives

$$\int_{T_F} \lambda_a \,\mathrm{d}x = \frac{|T_F|}{3}, \qquad \frac{\int_{T_F} \lambda_a x \,\mathrm{d}x}{\int_{T_F} \lambda_a \,\mathrm{d}x} = \frac{2a + b_F + c_F}{4} = \frac{a + m_F}{2} =: z_F.$$

Since s is affine, the moment condition reads $0 = (|T_F|/3)s(z_F)$. The four points z_F are the images of the four side midpoints under the same invertible affine map $x \mapsto (a+x)/2$. These side midpoints are not collinear, so s has three noncollinear zeros and is identically zero. Thus the moment map is onto. Finally, the normalized aspect ratios and the admissible apex locations form a compact family without collapsed cones; the normalized moment matrix varies continuously and has full row rank at every configuration just considered. Its smallest row singular value has a uniform positive lower bound, which is exactly (3.4). ∎

COROLLARY 3.8 (Convex polygons for $k = 2$, any admissible apex). *For $d = 2$ and $k = 2$, the original local bubble-moment condition holds uniformly on any shape-regular family of convex polygons satisfying Section 2.1, for* every *admissible cone apex.*

Proof. Every nonrectangular convex polygon has two edge directions that are neither parallel nor perpendicular, and Lemma 3.4 proves injectivity there. If a convex polygon has no such pair, it is a rectangle after merging collinear sides, and Proposition 3.7 applies. The argument for subdivided rectangle sides is the summation of the moment conditions on each subcone, as in the proof of Theorem 3.10. Uniformity follows by normalizing diameter, taking the compact closure of admissible vertex-and-apex configurations with bounded edge count and nondegenerate cones, and using continuity of the smallest row singular value. Unlike the case of general $k \geq 3$, every off-center rectangle in this closure is covered by Proposition 3.7. ∎

LEMMA 3.9 (Uniform interiority of the centroid). *Let $K \subset \mathbb{R}^2$ be convex, with diameter h_K, and suppose K contains a disk $B_{\rho h_K}(\boldsymbol{z})$. Its area centroid $\boldsymbol{g}_K = |K|^{-1} \int_K \boldsymbol{x} \, \mathrm{d}\boldsymbol{x}$ satisfies*

$$\operatorname{dist}(\boldsymbol{g}_K, \partial K) \geq \frac{\rho^3}{8} h_K.$$

In particular, $\boldsymbol{g}_K$ is an admissible uniformly interior cone apex. Under the face-size assumptions of Section 2.1, its face cones remain uniformly shape regular.

Proof. For a side F, let $d_F(\boldsymbol{x})$ be the signed Euclidean distance to its supporting line, positive on K. Convexity gives $d_F \geq 0$ on K and $d_F(\boldsymbol{z}) \geq \rho h_K$. On $B_{\rho h_K/2}(\boldsymbol{z})$ we therefore have $d_F \geq \rho h_K/2$. Since d_F is affine and $|K| \leq \pi h_K^2$,

$$d_F(\boldsymbol{g}_K) = \frac{1}{|K|} \int_K d_F(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} \geq \frac{\rho h_K}{2} \frac{\pi(\rho h_K/2)^2}{\pi h_K^2} = \frac{\rho^3}{8} h_K.$$

This holds for every supporting side. Their half-plane intersection is K, so the disk of this radius about $\boldsymbol{g}_K$ lies in K. The stated cone regularity then follows from the uniform face-length lower bound and the diameter bound. ∎

THEOREM 3.10 (Two-dimensional convex polygons with a canonical apex). *Let $k \geq 2$ and let $K \subset \mathbb{R}^2$ be a convex polygon satisfying the mesh-regularity assumptions of Section 2.1. Subdivide K into cones $T_F = \operatorname{conv}(\boldsymbol{g}_K, F)$, where $\boldsymbol{g}_K = |K|^{-1} \int_K \boldsymbol{x} \, \mathrm{d}x$ is its centroid. Then, with $\mathbb{B}_k(\operatorname{div}, K; \mathbb{S})$ defined by (3.3),*

$$\inf_{0 \neq \boldsymbol{q} \in \mathbb{P}_{k-1}(K;\mathbb{S})} \sup_{0 \neq \boldsymbol{\tau} \in \mathbb{B}_k(\operatorname{div},K;\mathbb{S})} \frac{(\boldsymbol{\tau}, \boldsymbol{q})_K}{\|\boldsymbol{\tau}\|_K \, \|\boldsymbol{q}\|_K} \geq \beta_* > 0, \tag{3.8}$$

where β_* depends only on k and the mesh-regularity constants.

Proof. Step 1: elementwise injectivity. Let $m = k - 1 \geq 1$, and let $\boldsymbol{q} \in \mathbb{P}_m(K; \mathbb{S})$ annihilate every bubble. On each cone T_F, testing its normal–normal component against $\lambda_{\boldsymbol{g}_K}(\boldsymbol{n}_F^T \boldsymbol{q} \boldsymbol{n}_F)$ gives

$$\int_{T_F} \lambda_{\boldsymbol{g}_K} |\boldsymbol{n}_F^T \boldsymbol{q} \boldsymbol{n}_F|^2 = 0.$$

Thus $\boldsymbol{n}_F^T \boldsymbol{q} \boldsymbol{n}_F = 0$ on T_F, and polynomial continuation makes it zero on K. Testing the tangential–tangential bubble against $(\boldsymbol{t}_F^T \boldsymbol{q} \boldsymbol{t}_F)\boldsymbol{t}_F \otimes \boldsymbol{t}_F$ likewise gives $\boldsymbol{t}_F^T \boldsymbol{q} \boldsymbol{t}_F = 0$ on K. Hence $\boldsymbol{q} = s_F \operatorname{sym}(\boldsymbol{n}_F \otimes \boldsymbol{t}_F)$ for a scalar polynomial $s_F \in \mathbb{P}_m(K)$.

For two edges F, G whose directions make an angle θ neither zero nor $\pi/2$ modulo π, their shear-tensor lines are distinct: indeed the components of $\operatorname{sym}(\boldsymbol{n}_F \otimes \boldsymbol{t}_F)$ in the two-dimensional traceless symmetric matrix space rotate through 2θ. The two one-dimensional lines intersect only at zero; therefore $\boldsymbol{q} = 0$.

Suppose no such pair of edges exists. Choosing one edge direction, all remaining sides must be parallel to it or to its perpendicular. After a rigid rotation the polygon has horizontal and vertical sides; convexity forces it to be a rectangle, possibly with collinear subdivisions of its sides. The centroid of the rectangle is its center. For a subdivided side the apex weight is the same affine function on each subcone because the corresponding supporting line is the same. Summing the orthogonality relations on those subcones gives the moment relation on the cone over the full side. The centered-rectangle theorem, Theorem 3.6, therefore implies $\boldsymbol{q} = 0$ here too. Thus the adjoint of the moment operator is injective on every admissible polygon.

Step 2: a uniform inf–sup constant. Translate the centroid to the origin, rotate if convenient, and normalize the diameter to one. Lemma 3.9 ensures that the centroid remains a definite distance from every supporting edge. Uniformly bounded edge count and the lower edge-length bound make this family precompact in the space of ordered vertex configurations. Its closure contains only nondegenerate convex polygons, possibly with collinear consecutive edges, and each limit still has its apex at the polygon centroid. There are finitely many possible edge counts.

For each edge count, pull the piecewise polynomial bubble bases back from fixed reference triangles, and use continuously normalized Gram matrices for the L^2 norms of the bubble and polynomial spaces. Because all cones remain nondegenerate, the normalized moment matrix depends continuously on the vertices. Its smallest *row* singular value is positive at each limiting geometry by Step 1, including centered rectangles. It therefore has a strictly positive minimum over the compact closure. The minimum across the finitely many edge counts supplies a number $\beta_* > 0$ depending only on k and the mesh-regularity parameters. Undoing the diameter scaling gives (3.4) uniformly. ∎

Remark 3.11 (An arbitrary off-center rectangle apex). Theorem 3.6 uses the center of each rectangle; the affine scaling in its proof does not send a general interior apex to the square center. Thus the proof above settles the convex-polygon family with the explicit centroid choice, not the question of unisolvence of the original $k \geq 2$ bubble space for every admissible *off-center* rectangular cone decomposition. For a general star-center choice one may retain Assumption 3.1. This does not restrict the centroid-based method on convex polygons.

COROLLARY 3.12 (Lowest order on convex polygons). *Let $d = 2$ and $k = 1$. On every individual convex nonrectangular polygon, the original unenriched bubble moment operator is onto; a uniform inf–sup bound requires an angular-separation condition or the lowest-order enrichment. On rectangles it fails by Remark 3.14.*

Proof. A convex nonrectangular polygon must contain two face directions that are neither parallel nor orthogonal; otherwise all of its sides are parallel to two orthogonal directions and the polygon is a rectangle. Apply Lemma 3.4. The rectangular failure is stated in Remark 3.14. ∎

Remark 3.13 (General star-shaped polygons). The argument of Lemma 3.4 does not use convexity. Therefore the intrinsic moment operator is onto, for every $k \geq 1$, on any star-shaped polygon having two face directions that are neither parallel nor orthogonal. Uniformity without the enrichment needs a uniform nondegeneracy condition on those directions. The only two-dimensional star-shaped geometries not covered by this argument or by Theorem 3.6 are nonconvex orthogonal polygons, whose sides all lie in two mutually orthogonal directions. We do not need such elements for Theorem 3.10, and we leave their purely local moment characterization separate rather than imposing an artificial geometric spanning hypothesis.

Remark 3.14 (Why the unenriched $k = 1$ space is different on rectangles). For $k = 1$ we have $m = 0$ and the normal–tangential bubble block in (3.3) is absent. The constant shear tensor $\mathrm{sym}(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2)$ is orthogonal to the remaining normal–normal and tangential–tangential bubble blocks on an axis-aligned rectangle. Hence the local moment operator is not onto and Assumption 3.1 fails. This is an algebraic distinction, not merely a limitation of the proof.

Lemma 3.15 (Local rectangular shear kernel). *Let K be a two-dimensional rectangle with center $\boldsymbol{x}_K$, and let the unenriched $k = 1$ space* (3.1) *be used. Set*

$$\boldsymbol{S} := \boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_1, \qquad \boldsymbol{w}_K(\boldsymbol{x}) := \boldsymbol{S}(\boldsymbol{x} - \boldsymbol{x}_K).$$

For the local velocity pair $\boldsymbol{v}_K = \{\boldsymbol{w}_K, \boldsymbol{0}\}$, one has

$$\varepsilon_K^w(\boldsymbol{v}_K) = \boldsymbol{0}, \qquad \|\boldsymbol{v}_K\|_{1,h,K}^2 \simeq |K|.$$

Proof. Since $\boldsymbol{w}_K \in \mathbb{P}_1(K;\mathbb{R}^2)$, $\boldsymbol{\Pi}_K^o \boldsymbol{w}_K = \boldsymbol{w}_K$ and $\varepsilon(\boldsymbol{w}_K) = \boldsymbol{S}$. Moreover the normal jump vanishes and the projected tangential jump on a face F is $-\Pi_F \boldsymbol{w}_K$.

Consider one cone T with base face F, outward normal $\boldsymbol{n}$, unit tangent $\boldsymbol{t}$, and height d_F from $\boldsymbol{x}_K$ to F. For the normal–normal block $p\,\boldsymbol{n} \otimes \boldsymbol{n}$, $p \in \mathbb{P}_1(T)$, both the volume term and the normal boundary term vanish because $\boldsymbol{n}^T \boldsymbol{S} \boldsymbol{n} = 0$. For the purely tangential block $c\,\boldsymbol{t} \otimes \boldsymbol{t}$, the traction on F vanishes and the volume term is zero because $\boldsymbol{t}^T \boldsymbol{S} \boldsymbol{t} = 0$.

It remains to test with $\boldsymbol{\sigma} = c\ \mathrm{sym}(\boldsymbol{n} \otimes \boldsymbol{t})$. Since $\boldsymbol{\sigma}\boldsymbol{n} = (c/2)\boldsymbol{t}$ and $\boldsymbol{w}_K \cdot \boldsymbol{t} = d_F\, \boldsymbol{n}^T \boldsymbol{S} \boldsymbol{t}$ on F, while $|T| = |F| d_F/2$, we obtain

$$(\boldsymbol{S}, \boldsymbol{\sigma})_T + \langle -\Pi_F \boldsymbol{w}_K, \boldsymbol{\sigma}\boldsymbol{n} \rangle_F = c|T|\, \boldsymbol{n}^T \boldsymbol{S} \boldsymbol{t} - \frac{c}{2} |F| d_F\, \boldsymbol{n}^T \boldsymbol{S} \boldsymbol{t} = 0.$$

The stress space is the direct sum of these three blocks on every cone, so (3.14) gives $\varepsilon_K^w(\boldsymbol{v}_K) = 0$. The local mesh-dependent norm contains $\|\boldsymbol{S}\|_K^2 \simeq |K|$ and is therefore comparable to $|K|$. ∎

The local degeneracy cannot be controlled solely through the global boundary condition. In fact, the local shear kernel can generate a boundary-layer near-kernel on a sequence of uniformly refined rectangular meshes. The next proposition records this point; it motivates a minimal local enrichment at lowest order.

Proposition 3.16 (Boundary-layer shear mode for the unenriched $k = 1$ space). *Let $\Omega = (0,1)^2$ be partitioned into a uniform $N \times N$ rectangular mesh, $h = N^{-1}$,*

and use the unenriched $k = 1$ stress space (3.1)*. Then there is a sequence $\boldsymbol{v}_h \in \boldsymbol{V}_h$ such that*

$$\|\boldsymbol{v}_h\|_{1,h} \gtrsim 1, \qquad \|\varepsilon_h^w(\boldsymbol{v}_h)\|^2 \lesssim h.$$

Consequently, a mesh-independent estimate $\|\boldsymbol{v}_h\|_{1,h} \lesssim \|\varepsilon_h^w(\boldsymbol{v}_h)\|$ cannot hold for the unenriched lowest-order rectangular method.

Proof. Let

$$\boldsymbol{S} := \boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_1.$$

On every element K that does not touch $\partial\Omega$, set

$$\boldsymbol{w}_K(\boldsymbol{x}) := \boldsymbol{S}(\boldsymbol{x} - \boldsymbol{x}_K), \qquad \boldsymbol{v}_h^{\mathrm{div}}|_K := \boldsymbol{w}_K, \qquad \boldsymbol{v}_h^t|_{\partial K} := 0.$$

For two neighboring interior rectangles, the normal traces of these local polynomials coincide on their common face; the tangential variable is already single-valued. Lemma 3.15 gives $\varepsilon_K^w(\boldsymbol{v}_h) = 0$ on every such interior element, while $\|\boldsymbol{v}_h\|_{1,h,K}^2 \simeq |K|$.

On the one-cell boundary layer, prescribe zero normal trace on the exterior faces and the preceding interior normal trace on faces shared with an interior cell, keep the tangential variable zero, and choose the remaining local VEM DoFs with uniformly bounded reference-element size. Unisolvence of $\boldsymbol{V}_K$ gives such local functions. The prescribed traces have amplitude $O(h)$, and scaling from the reference rectangle therefore gives

$$\|\boldsymbol{v}_h\|_{1,h,K}^2 + \|\varepsilon_K^w(\boldsymbol{v}_h)\|_K^2 \lesssim h^2$$

on each boundary-layer cell. There are $O(h^{-1})$ such cells. The interior cells occupy a region of area bounded below independently of h, so summing gives

$$\|\boldsymbol{v}_h\|_{1,h}^2 \gtrsim 1, \qquad \|\varepsilon_h^w(\boldsymbol{v}_h)\|^2 \lesssim h. \qquad \square$$

3.2. A minimal lowest-order enrichment. For the lowest-order method, we use the following local symmetric-tensor space. For $k = 1$ and for a cone T with base face F, define

$$\widetilde{\Sigma}_1(T) := \Sigma_1(T) \oplus \lambda_{\boldsymbol{x}_K}\, \mathrm{sym}(\mathscr{T}^F \otimes \boldsymbol{n}_F). \tag{3.9}$$

The added functions vanish in normal traction on the primal face F because $\lambda_{\boldsymbol{x}_K}|_F = 0$. Hence the enrichment introduces no new globally coupled trace unknowns; it changes only the local symmetric-stress reconstruction. Let $\widetilde{\Sigma}_h^{-1}(K)$ and $\widetilde{\mathbb{B}}_1(\mathrm{div}, K; \mathbb{S})$ denote the corresponding broken local space and zero-traction bubble space.

THEOREM 3.17 (Intrinsic stability of the lowest-order symmetric-tensor space). *Let K satisfy the mesh regularity assumptions of Section 2. For $k = 1$, replace $\Sigma_1(T)$ by $\widetilde{\Sigma}_1(T)$ in* (3.9)*. Then*

$$\inf_{0 \neq \boldsymbol{q} \in \mathbb{P}_0(K;\mathbb{S})} \sup_{0 \neq \boldsymbol{\tau} \in \widetilde{\mathbb{B}}_1(\mathrm{div},K;\mathbb{S})} \frac{(\boldsymbol{\tau}, \boldsymbol{q})_K}{\|\boldsymbol{\tau}\|_K \, \|\boldsymbol{q}\|_K} \geq \widetilde{\beta}_* > 0, \tag{3.10}$$

where $\widetilde{\beta}_*$ depends only on the mesh-regularity constants. In particular, the result includes two-dimensional rectangles and parallelograms.

Proof. Fix one cone $T \subset K$ with base face F. Relative to the orthogonal normal–tangential decomposition associated with F, every constant symmetric tensor has the unique decomposition

$$\mathbb{S} = \mathrm{span}\{\boldsymbol{n}_F \otimes \boldsymbol{n}_F\} \oplus \mathrm{sym}(\mathscr{T}^F \otimes \boldsymbol{n}_F) \oplus \mathscr{T}^F(\mathbb{S}).$$

The enriched bubble space contains, with support in this cone,

$$\lambda_{\boldsymbol{x}_K}(\boldsymbol{n}_F \otimes \boldsymbol{n}_F), \qquad \lambda_{\boldsymbol{x}_K}\operatorname{sym}(\mathscr{T}^F \otimes \boldsymbol{n}_F), \qquad \mathscr{T}^F(\mathbb{S}).$$

Let $\boldsymbol{q} \in \mathbb{P}_0(K;\mathbb{S})$ be orthogonal to $\widetilde{\mathbb{B}}_1(\operatorname{div}, K;\mathbb{S})$. Testing with the three blocks above yields respectively the normal–normal, normal–tangential, and purely tangential components of $\boldsymbol{q}$ equal to zero, because $\int_T \lambda_{\boldsymbol{x}_K} > 0$ and $|T| > 0$. Thus $\boldsymbol{q} = 0$, so the moment operator is onto. After scaling to unit diameter, the three block moments form a uniformly nonsingular finite-dimensional system; shape regularity of the cone gives a lower singular-value bound depending only on the mesh-regularity constants. Scaling back proves (3.10). ∎

Remark 3.18 (Unified interpretation). The enrichment (3.9) is precisely the missing normal–tangential zero-traction block at lowest order. For $k \geq 2$ that role is played by $\lambda_{\boldsymbol{x}_K}\operatorname{sym}(\mathbb{P}_{k-2}(T;\mathscr{T}^F) \otimes \boldsymbol{n}_F)$. Thus the enriched $k = 1$ construction is not a penalty or an extrinsic stabilization; it is a minimal local completion of the symmetric-stress bubble mechanism.

Henceforth, when $k = 1$ we use the enriched local space $\widetilde{\Sigma}_1(T)$ from (3.9); for $k \geq 2$ we use $\Sigma_k(T)$ from (3.1). To avoid a separate notation in the remainder of the paper, both cases are denoted by $\Sigma_k(T)$, $\Sigma_h^{-1}(K)$, and $\mathbb{B}_k(\operatorname{div}, K;\mathbb{S})$. With this convention, Assumption 3.1 is proved for $k = 1$ by Theorem 3.17, and for two-dimensional shape-regular convex polygons with $k \geq 2$ by Theorem 3.10.

LEMMA 3.19. *Assume Assumption 3.1. For $K \in \mathcal{K}_h$, the space $\Sigma_h^{-1}(K)$ is uniquely determined by the following DoFs:*

$$\int_F \boldsymbol{\sigma}\boldsymbol{n} \cdot \boldsymbol{q}\, \mathrm{d}s, \quad \boldsymbol{q} \in V_k(F), \quad F \subset \partial K, \tag{3.11a}$$

$$\int_K \boldsymbol{\sigma} : \boldsymbol{\tau}\, \mathrm{d}x, \quad \boldsymbol{\tau} \in \mathcal{Z}_K, \tag{3.11b}$$

$$\int_K \boldsymbol{\sigma} : \boldsymbol{q}\, \mathrm{d}x, \quad \boldsymbol{q} \in \mathbb{P}_{k-1}(K;\mathbb{S}). \tag{3.11c}$$

Proof. Suppose that all moments vanish. The face moments imply $\boldsymbol{\sigma}\boldsymbol{n} = 0$ on ∂K, hence $\boldsymbol{\sigma} \in \mathbb{B}_k(\operatorname{div}, K;\mathbb{S})$. The moments (3.11b) imply $\boldsymbol{\sigma} \in \mathcal{E}_K$, whereas (3.11c) gives $\mathcal{M}_K\boldsymbol{\sigma} = 0$, i.e., $\boldsymbol{\sigma} \in \mathcal{Z}_K$. Thus $\boldsymbol{\sigma} \in \mathcal{E}_K \cap \mathcal{Z}_K = \{0\}$. The trace map is onto $\bigoplus_{F\subset\partial K} V_k(F)$ and has the bubble space as its kernel. Since $\dim \mathcal{E}_K = \dim \mathbb{P}_{k-1}(K;\mathbb{S})$, the number of moments in (3.11) equals $\dim \Sigma_h^{-1}(K)$. Hence the DoFs are unisolvent. ∎

Remark 3.20 (Cost of the lowest-order enrichment). For $k = 1$, the original local symmetric-tensor space has five polynomial coefficients per cone, whereas the enriched space (3.9) has six. Thus the dimension of the polygon-local stress space increases from $5N_K$ to $6N_K$, with precisely one additional normal–tangential bubble mode on each cone T_F. Since this mode has zero traction on the outer edge F, the number of outer-edge stress traction moments remains $3N_K$. The enrichment changes neither the velocity nor the pressure degrees of freedom and introduces no globally coupled stress unknowns. Moreover, because the tensor space is broken across cones, its L^2 mass matrix is block diagonal with respect to the cone subdivision: for $k = 1$, the cone blocks increase only from size 5×5 to 6×6.

LEMMA 3.21 (Stable local lifting). *Assume Assumption 3.1. Given $\boldsymbol{g}_K \in \mathbb{P}_{k-1}(K;\mathbb{S})$ and face data $\boldsymbol{r}_F \in V_k(F)$, there is a unique $\boldsymbol{\sigma}_K \in \Sigma_h^{-1}(K)$ whose polynomial mo-*

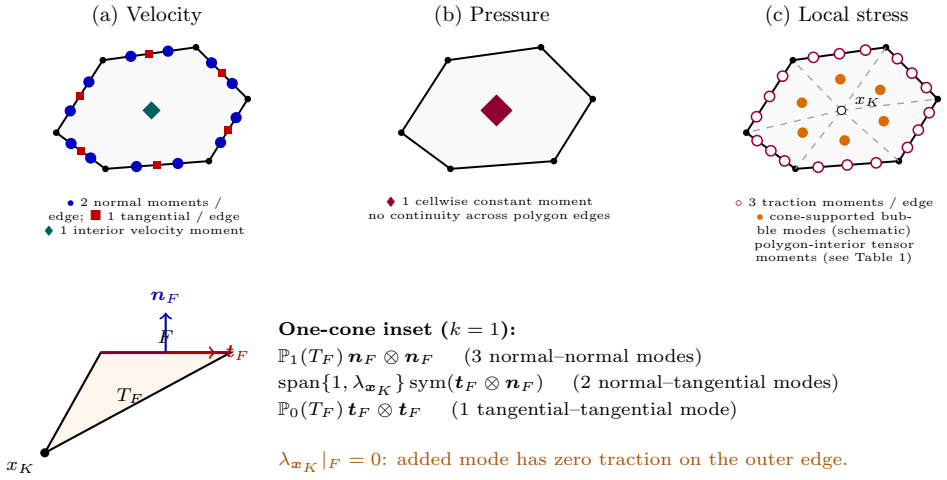


FIG. 1. *Degrees of freedom and local reconstruction for $k = 1$ on a hexagonal polygon K. (a) The $H(\mathrm{div})$ velocity has two normal moments per edge and one interior moment; the separate tangential trace has one moment per edge. (b) The pressure has one cellwise constant moment. (c) The enriched symmetric-tensor space is broken over the six cones $T_F = \mathrm{conv}(x_K, F)$, where x_K is the centroid. Each outer edge has three local traction moments (two normal and one tangential). The orange symbols indicate cone-supported bubble modes schematically, not individual degrees of freedom. The inset displays the six local tensor modes on one cone, including the added normal–tangential bubble. The remaining interior tensor degrees of freedom are the three constant symmetric-tensor moments and $\dim \mathcal{Z}_K = 3N_K - 3$ kernel moments, where N_K is the number of edges. Stress traction is not constrained to be continuous across radial cone interfaces or between adjacent polygons, and the stress reconstruction adds no globally coupled unknowns.*

TABLE 1

Local degrees of freedom for $k = 1$ on a polygon K with N_K edges. The enriched symmetric-stress space has six local coefficients per cone, but introduces no globally coupled stress unknowns.

Variable	DoF family	General N_K	$N_K = 6$
Velocity	Edge normal moments	$2N_K$	12
	Edge tangential moments	N_K	6
	Interior Type III moment	1	1
	Total	$3N_K + 1$	19
Pressure	Interior constant moment	1	1
Stress	Outer-edge traction moments	$3N_K$	18
	Interior constant tensor moments	3	3
	Interior kernel moments ($\mathcal{Z}_K$)	$3N_K - 3$	15
	Total (local)	$6N_K$	36

ments equal those of $\boldsymbol{g}_K$, whose traction moments equal those of $\boldsymbol{r}_F$, and whose $\mathcal{Z}_K$ moments vanish. Moreover,

$$\|\boldsymbol{\sigma}_K\|_K \lesssim \|\boldsymbol{g}_K\|_K + \sum_{F \subset \partial K} h_K^{1/2} \|\boldsymbol{r}_F\|_F. \tag{3.12}$$

Proof. We construct the lifting in two stages, to exhibit the uniform constant. Decompose each prescribed face vector as $\boldsymbol{r}_F = r_F^n \boldsymbol{n}_F + \boldsymbol{r}_F^t$. Extend its polynomial

normal and tangential components from F to T_F by keeping them constant on lines normal to F; denote these extensions by $\widehat{r}_F^n$ and $\widehat{\boldsymbol{r}}_F^t$. By the definition of $\Sigma_k(T_F)$, the tensor

$$\boldsymbol{\sigma}^{\mathrm{tr}}|_{T_F} = \widehat{r}_F^n \boldsymbol{n}_F \otimes \boldsymbol{n}_F + 2\,\mathrm{sym}(\boldsymbol{n}_F \otimes \widehat{\boldsymbol{r}}_F^t)$$

belongs to the local stress space and has exactly the prescribed traction $\boldsymbol{r}_F$ on the base face F. The other cones are independently treated in the same way. Fixed-degree polynomial scaling on the uniformly regular cones yields

$$\|\boldsymbol{\sigma}^{\mathrm{tr}}\|_K \lesssim \Big(\sum_{F \subset \partial K} h_K \|\boldsymbol{r}_F\|_F^2 \Big)^{1/2}. \tag{3.13}$$

Let $P_{\mathcal{Z}_K}$ denote the $L^2(K)$-orthogonal projector onto $\mathcal{Z}_K = \ker \mathcal{M}_K$ and define $\boldsymbol{\sigma}^0 = \boldsymbol{\sigma}^{\mathrm{tr}} - P_{\mathcal{Z}_K}\boldsymbol{\sigma}^{\mathrm{tr}}$. Elements of $\mathcal{Z}_K$ are traction-free bubbles, so this subtraction leaves all prescribed face moments unchanged. Furthermore $\boldsymbol{\sigma}^0 \perp \mathcal{Z}_K$ and $\|\boldsymbol{\sigma}^0\|_K \le 2\|\boldsymbol{\sigma}^{\mathrm{tr}}\|_K$.

The moment residual on $Q_K = \mathbb{P}_{k-1}(K;\mathbb{S})$ is the linear functional

$$L(\boldsymbol{q}) = (\boldsymbol{g}_K, \boldsymbol{q})_K - (\boldsymbol{\sigma}^0, \boldsymbol{q})_K, \qquad \boldsymbol{q} \in Q_K.$$

Its dual L^2 norm is bounded by $\|L\|_{Q_K^*} \le \|\boldsymbol{g}_K\|_K + \|\boldsymbol{\sigma}^0\|_K$. The intrinsic inf–sup condition is equivalent to the existence of a unique $\boldsymbol{\sigma}^b \in \mathcal{E}_K$ for which

$$(\boldsymbol{\sigma}^b, \boldsymbol{q})_K = L(\boldsymbol{q}) \quad (\boldsymbol{q} \in Q_K), \qquad \|\boldsymbol{\sigma}^b\|_K \le \beta_*^{-1}\|L\|_{Q_K^*}.$$

Finally set $\boldsymbol{\sigma}_K = \boldsymbol{\sigma}^0 + \boldsymbol{\sigma}^b$. Its face tractions are $\boldsymbol{r}_F$; its Q_K moments equal those of $\boldsymbol{g}_K$; and both terms are orthogonal to $\mathcal{Z}_K$. The estimates above imply

$$\|\boldsymbol{\sigma}_K\|_K \le (1+\beta_*^{-1})\|\boldsymbol{\sigma}^0\|_K + \beta_*^{-1}\|\boldsymbol{g}_K\|_K \lesssim \|\boldsymbol{g}_K\|_K + \Big(\sum_F h_K \|\boldsymbol{r}_F\|_F^2\Big)^{1/2},$$

which is stronger than (3.12). Existence follows from this construction and uniqueness from Lemma 3.19. ∎

Let

$$\Sigma_h^{-1} = \{\boldsymbol{\sigma} \in L^2(\Omega;\mathbb{S}) : \boldsymbol{\sigma}|_T \in \Sigma_k(T), T \in \mathcal{T}_h\}.$$

3.3. Weak strain. For any $\boldsymbol{v} = \{\boldsymbol{v}^{\mathrm{div}}, \boldsymbol{v}^t\} \in \boldsymbol{V}_h$, we define $\varepsilon_h^w(\boldsymbol{v}) \in \Sigma_h^{-1}$ satisfying $\varepsilon_h^w(\boldsymbol{v})|_K = \varepsilon_K^w(\boldsymbol{v})$ and

$$\begin{aligned}(\varepsilon_K^w(\boldsymbol{v}), \boldsymbol{\sigma})_K = {}& (\varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{v}^{\mathrm{div}}), \boldsymbol{\sigma})_K \\ & + \langle (\boldsymbol{v}^{\mathrm{div}} - \boldsymbol{\Pi}_K^o \boldsymbol{v}^{\mathrm{div}}) \cdot \boldsymbol{n}, (\boldsymbol{\sigma}\boldsymbol{n}) \cdot \boldsymbol{n} \rangle_{\partial K} \\ & + \langle \boldsymbol{v}^t - \Pi_F \boldsymbol{\Pi}_K^o \boldsymbol{v}^{\mathrm{div}}, \Pi_F(\boldsymbol{\sigma}\boldsymbol{n}) \rangle_{\partial K}\end{aligned} \tag{3.14}$$

for all $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$, with Π_F applied facewise. Since $\boldsymbol{\sigma}$ is discontinuous inside K, integration by parts must include the internal jumps of $\boldsymbol{\sigma}\boldsymbol{n}$. Dropping these terms does not give an equivalent definition.

The normal and tangential boundary jumps are

$$\begin{aligned}\mathcal{J}_{\partial K}^{\boldsymbol{n}}(\boldsymbol{v}) &:= ((\boldsymbol{v}^{\mathrm{div}} - \boldsymbol{\Pi}_K^o \boldsymbol{v}^{\mathrm{div}}) \cdot \boldsymbol{n})\boldsymbol{n}, \\ \mathcal{J}_{\partial K}^{\boldsymbol{\tau}}(\boldsymbol{v}) &:= \boldsymbol{v}^t - \Pi_F \boldsymbol{\Pi}_K^o \boldsymbol{v}^{\mathrm{div}}.\end{aligned}$$

Their sum is $\mathcal{J}_{\partial K}(\boldsymbol{v}) := \mathcal{J}^{\boldsymbol{n}}_{\partial K}(\boldsymbol{v}) + \mathcal{J}^{\boldsymbol{\tau}}_{\partial K}(\boldsymbol{v})$.

On the face $F \subset \partial K$, we define $(\boldsymbol{\Pi}^t_{\partial K}\boldsymbol{v})|_F \in \boldsymbol{V}_F$ as

$$\int_F \boldsymbol{\Pi}^t_{\partial K}\boldsymbol{v} \cdot \boldsymbol{q}\,\mathrm{d}s = \int_F \boldsymbol{v}|_K \cdot \boldsymbol{q}\,\mathrm{d}s \quad \forall \boldsymbol{q} \in \boldsymbol{V}_F.$$

Define the notation $\boldsymbol{\Pi}_{\partial K}$ as

$$\boldsymbol{\Pi}_{\partial K}\boldsymbol{v} := (\boldsymbol{v}\cdot\boldsymbol{n})\boldsymbol{n} + \boldsymbol{\Pi}^t_{\partial K}\boldsymbol{v}.$$

Since $\boldsymbol{\sigma}\boldsymbol{n}|_F \in V_k(F)$, (3.14) becomes

$$(\varepsilon^w_K(\boldsymbol{v}), \boldsymbol{\sigma})_K = (\varepsilon(\boldsymbol{\Pi}^o_K \boldsymbol{v}^{\mathrm{div}}), \boldsymbol{\sigma})_K + \langle \boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v}), \boldsymbol{\sigma}\boldsymbol{n}\rangle_{\partial K}.$$

For $\boldsymbol{v}_h \in \boldsymbol{V}_h$, define the mesh-dependent seminorm

$$\|\boldsymbol{v}_h\|^2_{1,h} := \sum_{K\in\mathcal{K}_h} \|\boldsymbol{v}_h\|^2_{1,h,K},$$

where the seminorm on each element is defined as

$$\|\boldsymbol{v}_h\|^2_{1,h,K} := \|\varepsilon(\boldsymbol{\Pi}^o_K \boldsymbol{v}^{\mathrm{div}}_h)\|^2_K + h_K^{-1}\|\boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v}_h)\|^2_{\partial K}.$$

LEMMA 3.22. *The mesh-dependent quantity $\|\boldsymbol{v}_h\|_{1,h}$ defines a norm on the virtual element space $\boldsymbol{V}_h$.*

Proof. If $\|\boldsymbol{v}_h\|_{1,h} = 0$, then on each K, $\boldsymbol{\Pi}^o_K \boldsymbol{v}^{\mathrm{div}}_h = \boldsymbol{r}_K \in \mathbf{RM}(K)$, and

$$\boldsymbol{v}^{\mathrm{div}}_h \cdot \boldsymbol{n} = \boldsymbol{r}_K \cdot \boldsymbol{n}, \qquad \boldsymbol{v}^t_h = \Pi_F \boldsymbol{r}_K \quad \text{on each } F \subset \partial K.$$

The second identity uses $\Pi_F \boldsymbol{r}_K \in \mathbf{RM}(F)$. The normal component of $\boldsymbol{v}^{\mathrm{div}}_h$ and the single-valued tangential field $\boldsymbol{v}^t_h$ show that neighboring rigid motions agree on their common face. A rigid motion that vanishes on a face is zero, so connectedness gives one global rigid motion $\boldsymbol{r}$. The boundary conditions give both $\boldsymbol{r}\cdot\boldsymbol{n} = 0$ and $\Pi_F \boldsymbol{r} = 0$ on boundary faces. Hence $\boldsymbol{r} = 0$. The normal face moments and all polynomial volume moments of $\boldsymbol{v}^{\mathrm{div}}_h$ now vanish. Unisolvence of $\boldsymbol{V}_K$ gives $\boldsymbol{v}^{\mathrm{div}}_h = 0$, and the tangential identity gives $\boldsymbol{v}^t_h = 0$. ∎

We also need control of the full jump. For $\boldsymbol{w} = \boldsymbol{\Pi}^o_K \boldsymbol{v}^{\mathrm{div}}_h$, Korn's inequality modulo rigid motions gives a $\boldsymbol{r}_K \in \mathbf{RM}(K)$ such that

$$h_K^{-1}\|\boldsymbol{w} - \boldsymbol{r}_K\|_K + \|\nabla(\boldsymbol{w} - \boldsymbol{r}_K)\|_K \lesssim \|\varepsilon(\boldsymbol{w})\|_K.$$

Since $\boldsymbol{\Pi}^t_{\partial K}$ reproduces $\Pi_F \boldsymbol{r}_K$, the trace inequality yields

$$h_K^{-1/2}\|\Pi_F \boldsymbol{w} - \boldsymbol{\Pi}^t_{\partial K}\boldsymbol{w}\|_{\partial K} \lesssim \|\varepsilon(\boldsymbol{w})\|_K.$$

The normal part of the jump is unchanged by $\boldsymbol{\Pi}_{\partial K}$. The unprojected tangential part has the same norm as the difference above. Thus

$$h_K^{-1/2}\|\mathcal{J}_{\partial K}(\boldsymbol{v}_h)\|_{\partial K} \lesssim \|\boldsymbol{v}_h\|_{1,h,K}. \tag{3.15}$$

The jump of $\boldsymbol{w}$ across a primal face is the difference of its two full boundary jumps; on a boundary face, $\boldsymbol{w}$ is minus its full boundary jump. Applying the broken Korn inequality [5] on the simplicial refinement therefore gives the global equivalence

$$\|\boldsymbol{v}_h\|^2_{1,h} \simeq \sum_{K\in\mathcal{K}_h} \left(\|\nabla\boldsymbol{\Pi}^o_K \boldsymbol{v}^{\mathrm{div}}_h\|^2_K + h_K^{-1}\|\boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v}_h)\|^2_{\partial K}\right). \tag{3.16}$$

This equivalence is global; it does not hold on an individual element.

LEMMA 3.23. *We have the discrete inf-sup condition*

$$\|\boldsymbol{v}\|_{1,h} \lesssim \sup_{0\neq\boldsymbol{\sigma}_h\in\Sigma_h^{-1}} \frac{\sum_{K\in\mathcal{K}_h}(\varepsilon_K^w(\boldsymbol{v}),\boldsymbol{\sigma}_h)_K}{\|\boldsymbol{\sigma}_h\|_{0,h}} \quad \forall\ \boldsymbol{v}\in\boldsymbol{V}_h. \tag{3.17}$$

Proof. Here $\|\boldsymbol{\sigma}\|_{0,h}^2 := \sum_K \|\boldsymbol{\sigma}\|_K^2$. For fixed $\boldsymbol{v}\in\boldsymbol{V}_h$, use Lemma 3.19 to choose $\boldsymbol{\sigma}|_K \in \Sigma_h^{-1}(K)$ with

$$\begin{aligned}(\boldsymbol{\sigma},\boldsymbol{\tau})_K &= (\varepsilon(\boldsymbol{\Pi}_K^o\boldsymbol{v}^{\mathrm{div}}),\boldsymbol{\tau})_K && \forall\boldsymbol{\tau}\in\mathbb{P}_{k-1}(K;\mathbb{S}),\\ \langle\boldsymbol{\sigma}\boldsymbol{n},\boldsymbol{q}\rangle_F &= \langle h_K^{-1}\boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v}),\boldsymbol{q}\rangle_F && \forall\boldsymbol{q}\in V_k(F),\ F\subset\partial K,\end{aligned}$$

and with the moments (3.11b) set to zero. Face moments are prescribed separately on each element side, using its outward normal. Testing with the polynomial strain and projected jump gives

$$\sum_K(\varepsilon_K^w(\boldsymbol{v}),\boldsymbol{\sigma})_K = \|\boldsymbol{v}\|_{1,h}^2.$$

Applying the stable local lifting, Lemma 3.21, with the prescribed polynomial and traction moments gives

$$\begin{aligned}\|\boldsymbol{\sigma}\|_K &\lesssim \|\varepsilon(\boldsymbol{\Pi}_K^o\boldsymbol{v}^{\mathrm{div}})\|_K + h_K^{-1/2}\|\boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v})\|_{\partial K},\\ \|\boldsymbol{\sigma}\|_{0,h} &\lesssim \|\boldsymbol{v}\|_{1,h}.\end{aligned}$$

Substitute this tensor into the supremum in (3.17). The result follows for $\boldsymbol{v}\neq 0$; the zero case is immediate. ∎

LEMMA 3.24. *For every $\boldsymbol{v}_h\in\boldsymbol{V}_h$,*

$$\|\boldsymbol{v}_h\|_{1,h} \lesssim \|\varepsilon_h^w(\boldsymbol{v}_h)\|.$$

Proof. Apply Lemma 3.23 and the Cauchy–Schwarz inequality. ∎

4. Numerical scheme.

Let

$$S_h := \{q_h\in L_0^2(\Omega) : q_h|_K\in\mathbb{P}_{k-1}(K) \text{ for all } K\in\mathcal{K}_h\}.$$

The discrete formulation of (2.1) seeks $\boldsymbol{u}_h = \{\boldsymbol{u}_h^{\mathrm{div}},\boldsymbol{u}_h^t\}\in\boldsymbol{V}_h$ and $p_h\in S_h$ such that

$$\begin{cases} a_h(\boldsymbol{u}_h,\boldsymbol{v}_h)+b(\boldsymbol{v}_h,p_h) = (\boldsymbol{f},\boldsymbol{\Pi}_h^o\boldsymbol{v}_h^{\mathrm{div}}) & \forall\boldsymbol{v}_h\in\boldsymbol{V}_h,\\ b(\boldsymbol{u}_h,q_h) = 0 & \forall q_h\in S_h,\end{cases} \tag{4.1}$$

where

$$a_h(\boldsymbol{u}_h,\boldsymbol{v}_h) := (\varepsilon_h^w(\boldsymbol{u}_h),\varepsilon_h^w(\boldsymbol{v}_h)), \qquad b(\boldsymbol{v}_h,p_h) := -(\operatorname{div}\boldsymbol{v}_h^{\mathrm{div}},p_h).$$

Remark 4.1 (Gradient forces and pressure robustness). Although the discrete velocity is exactly divergence-free, this alone does not guarantee full pressure robustness, since the elementwise L^2 projection $\boldsymbol{\Pi}_h^o\boldsymbol{v}_h^{\mathrm{div}}$ used in the load term of (4.1) is generally neither divergence-free nor $H(\mathrm{div})$-conforming; see [8] for a related discussion and a divergence-preserving reconstruction.

Nevertheless, the method is invariant under polynomial gradient forces. More precisely, let $\phi \in H^1(\Omega)$ satisfy $\nabla\phi|_K \in \mathbb{P}_k(K;\mathbb{R}^2)$ for every $K \in \mathcal{T}_h$. Then

$$\begin{aligned}(\nabla\phi, \boldsymbol{\Pi}_h^o \boldsymbol{v}_h^{\mathrm{div}}) &= (\nabla\phi, \boldsymbol{v}_h^{\mathrm{div}}) \\ &= -(\phi, \operatorname{div} \boldsymbol{v}_h^{\mathrm{div}}) \\ &= b(\boldsymbol{v}_h, Q_h\phi),\end{aligned}$$

where Q_h denotes the L^2 projection onto S_h. Here we have used polynomial orthogonality, normal conformity, and the homogeneous normal boundary condition. Consequently, replacing $\boldsymbol{f}$ by $\boldsymbol{f} + \nabla\phi$ leaves the discrete velocity unchanged and shifts the discrete pressure from p_h to $p_h + Q_h\phi$.

4.1. Local viscous stress and equivalent formulation. The viscosity is normalized so that the physical viscous stress is $\boldsymbol{\tau} = \varepsilon(\boldsymbol{u})$ and the total Cauchy stress is $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\boldsymbol{I}$. For each element define the strongly symmetric discrete viscous stress $\boldsymbol{\tau}_h|_K \in \Sigma_h^{-1}(K)$ by

$$(\boldsymbol{\tau}_h, \boldsymbol{\eta}_h)_K = (\varepsilon_K^w(\boldsymbol{u}_h), \boldsymbol{\eta}_h)_K \qquad \forall \boldsymbol{\eta}_h \in \Sigma_h^{-1}(K). \tag{4.2}$$

The corresponding local mixed representation of (4.1) is: find $(\boldsymbol{\tau}_h, \boldsymbol{u}_h, p_h)$ such that

$$\begin{cases} (\boldsymbol{\tau}_h - \varepsilon_K^w(\boldsymbol{u}_h), \boldsymbol{\eta}_h)_K = 0 & \forall \boldsymbol{\eta}_h \in \Sigma_h^{-1}(K),\ K \in \mathcal{K}_h, \\ \displaystyle\sum_K (\boldsymbol{\tau}_h, \varepsilon_K^w(\boldsymbol{v}_h))_K + b(\boldsymbol{v}_h, p_h) = (\boldsymbol{f}, \boldsymbol{\Pi}_h^o \boldsymbol{v}_h^{\mathrm{div}}) & \forall \boldsymbol{v}_h \in \boldsymbol{V}_h, \\ b(\boldsymbol{u}_h, q_h) = 0 & \forall q_h \in S_h. \end{cases} \tag{4.3}$$

The first equation yields $\boldsymbol{\tau}_h = \varepsilon_h^w(\boldsymbol{u}_h)$; substitution gives precisely (4.1). Thus the stress is computed locally and adds no globally coupled unknowns. The following proposition records three structural properties of the reconstructed stress. The tensor $\boldsymbol{\sigma}_h = \boldsymbol{\tau}_h - p_h\boldsymbol{I}$ is symmetric on each cone, but neither traction continuity nor a strong local momentum balance is asserted.

PROPOSITION 4.2 (Symmetry, energy identity, and bubble orthogonality of the reconstructed stress). *Let $\boldsymbol{\tau}_h = \varepsilon_h^w(\boldsymbol{u}_h)$ and set $\boldsymbol{G}_K = \varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{u}_h^{\mathrm{div}})$. Then*

$$\begin{aligned} &\boldsymbol{\tau}_h|_K \in \Sigma_h^{-1}(K), \\ &a_h(\boldsymbol{u}_h, \boldsymbol{u}_h) = \sum_{K\in\mathcal{K}_h} \|\boldsymbol{\tau}_h\|_{0,K}^2, \\ &(\boldsymbol{\tau}_h - \boldsymbol{G}_K, \boldsymbol{\eta}_h)_K = 0 \qquad \forall \boldsymbol{\eta}_h \in \mathbb{B}_k(\operatorname{div}, K;\mathbb{S}). \end{aligned} \tag{4.4}$$

Here $\mathbb{B}_k(\operatorname{div}, K;\mathbb{S})$ is defined in (3.3).

Proof. The first assertion follows from the definition of the local symmetric-tensor space. The second follows from the definition of a_h and $\boldsymbol{\tau}_h = \varepsilon_h^w(\boldsymbol{u}_h)$. For $\boldsymbol{\eta}_h \in \mathbb{B}_k(\operatorname{div}, K;\mathbb{S})$, its normal traction vanishes on ∂K. Testing (3.14) with $\boldsymbol{\eta}_h$ therefore eliminates both boundary terms and gives $(\varepsilon_K^w(\boldsymbol{u}_h), \boldsymbol{\eta}_h)_K = (\boldsymbol{G}_K, \boldsymbol{\eta}_h)_K$, as claimed. ∎

The reconstructed total stress $\boldsymbol{\sigma}_h = \boldsymbol{\tau}_h - p_h\boldsymbol{I}$ is strongly symmetric elementwise. The reconstruction does not, in general, impose continuity of $\boldsymbol{\sigma}_h\boldsymbol{n}$ across element interfaces or strong elementwise momentum equilibrium.

LEMMA 4.3. *For every $\boldsymbol{v}_h = \{\boldsymbol{v}_h^{\mathrm{div}}, \boldsymbol{v}_h^t\} \in \boldsymbol{V}_h$, there holds*

$$\|\varepsilon_h^w(\boldsymbol{v}_h)\|^2 \lesssim \|\boldsymbol{v}_h\|_{1,h}^2.$$

Proof. Set $\boldsymbol{\sigma} = \varepsilon_K^w(\boldsymbol{v}_h)$ in (3.14). The Cauchy–Schwarz and polynomial trace inequalities give

$$\|\varepsilon_K^w(\boldsymbol{v}_h)\|_K^2 \lesssim \Big(\|\varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}})\|_K + h_K^{-1/2}\|\boldsymbol{\Pi}_{\partial K}\mathcal{J}_{\partial K}(\boldsymbol{v}_h)\|_{\partial K}\Big)\|\varepsilon_K^w(\boldsymbol{v}_h)\|_K.$$

Divide by the strain norm when it is nonzero, square, and sum over K. ∎

LEMMA 4.4. *For $\boldsymbol{u}_h, \boldsymbol{v}_h \in \boldsymbol{V}_h$,*

$$|a_h(\boldsymbol{u}_h, \boldsymbol{v}_h)| \lesssim \|\boldsymbol{u}_h\|_{1,h}\|\boldsymbol{v}_h\|_{1,h},$$
$$a_h(\boldsymbol{u}_h, \boldsymbol{u}_h) \gtrsim \|\boldsymbol{u}_h\|_{1,h}^2.$$

LEMMA 4.5. *For $\boldsymbol{v}_h \in \boldsymbol{V}_h$ and $q_h \in S_h$,*

$$|b(\boldsymbol{v}_h, q_h)| \lesssim \|\boldsymbol{v}_h\|_{1,h}\|q_h\|.$$

Proof. Write $\boldsymbol{w} = \boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}}$. For $q \in \mathbb{P}_{k-1}(K)$, the volume projection and integration by parts give

$$(\operatorname{div} \boldsymbol{v}_h^{\mathrm{div}}, q)_K = (\operatorname{div} \boldsymbol{w}, q)_K + \langle \mathcal{J}_{\partial K}^{\boldsymbol{n}}(\boldsymbol{v}_h)\cdot\boldsymbol{n}, q\rangle_{\partial K}.$$

Use $q = \operatorname{div} \boldsymbol{v}_h^{\mathrm{div}}$, the polynomial trace inequality, and $|\operatorname{div} \boldsymbol{w}| \le \sqrt{d}\,|\varepsilon(\boldsymbol{w})|$. This gives $\|\operatorname{div} \boldsymbol{v}_h^{\mathrm{div}}\|_K \lesssim \|\boldsymbol{v}_h\|_{1,h,K}$. Sum over K and apply the Cauchy–Schwarz inequality. ∎

LEMMA 4.6. *For every $p_h \in S_h$, there is a $\boldsymbol{v}_h \in \boldsymbol{V}_h$ with*

$$\operatorname{div} \boldsymbol{v}_h^{\mathrm{div}} = p_h, \qquad \|\boldsymbol{v}_h\|_{1,h} \lesssim \|p_h\|.$$

4.2. Interpolation. For $\boldsymbol{v} \in \boldsymbol{H}^1(K)$, define $\boldsymbol{I}_K^{\mathrm{div}}\boldsymbol{v} \in \boldsymbol{V}_K$ by matching its Type I–III DoFs with those of $\boldsymbol{v}$. Integration by parts with $q \in \mathbb{P}_{k-1}(K)$ gives the commuting identity

$$\operatorname{div} \boldsymbol{I}_K^{\mathrm{div}}\boldsymbol{v} = Q_h(\operatorname{div} \boldsymbol{v})|_K, \tag{4.5}$$

where Q_h is the elementwise L^2 projection onto $\mathbb{P}_{k-1}$; for mean-zero functions its range is S_h. The interpolant reproduces $\mathbb{P}_k(K;\mathbb{R}^d)$. Scaled stability of its DoFs gives

$$\|\boldsymbol{I}_K^{\mathrm{div}}\boldsymbol{v}\|_K \lesssim \|\boldsymbol{v}\|_K + h_K|\boldsymbol{v}|_{1,K}. \tag{4.6}$$

Indeed, the scaled face moments are bounded by $h_K^{1/2}\|\boldsymbol{v}\cdot\boldsymbol{n}\|_{\partial K}$, and the normalized volume moments by $\|\boldsymbol{v}\|_K$. The trace inequality and uniform norm equivalence for the unisolvent DoFs prove (4.6). These arguments are the usual face-VEM interpolation estimates; see [2, 3] for the underlying spaces and related stability results. Subtracting a degree-k polynomial approximant in (4.6) yields

$$\|\boldsymbol{v} - \boldsymbol{I}_K^{\mathrm{div}}\boldsymbol{v}\|_K \lesssim h_K^{k+1}|\boldsymbol{v}|_{k+1,K}. \tag{4.7}$$

Proof of Lemma 4.6. The continuous divergence right inverse [4] provides $\boldsymbol{v} \in \boldsymbol{H}_0^1(\Omega)$ with $\operatorname{div} \boldsymbol{v} = p_h$ and $|\boldsymbol{v}|_1 \lesssim \|p_h\|$. Set $\boldsymbol{v}_h^{\mathrm{div}}|_K = \boldsymbol{I}_K^{\mathrm{div}}\boldsymbol{v}$ and $\boldsymbol{v}_h^t|_F = \boldsymbol{\Pi}_{\partial K}^t\boldsymbol{v}$. The traces are single-valued and zero on the boundary, and (4.5) gives the required

divergence. Let $\boldsymbol{w} = \boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{v}$. Subtracting the mean of $\boldsymbol{v}$ in (4.6), and using polynomial inverse estimates, gives

$$\|\nabla \boldsymbol{w}\|_K \lesssim |\boldsymbol{v}|_{1,K}, \qquad \|\boldsymbol{v} - \boldsymbol{w}\|_K \lesssim h_K |\boldsymbol{v}|_{1,K}.$$

The normal interpolant trace is the face L^2 projection of $\boldsymbol{v} \cdot \boldsymbol{n}$. Thus both parts of the projected boundary jump are bounded by $\|\boldsymbol{v} - \boldsymbol{w}\|_{\partial K}$. The trace inequality now gives $\|\boldsymbol{v}_h\|_{1,h,K} \lesssim |\boldsymbol{v}|_{1,K}$. Sum over K to conclude. ∎

Since b has a minus sign, use the negative of this lifting to obtain

$$\|q_h\| \lesssim \sup_{0 \neq \boldsymbol{v}_h \in \boldsymbol{V}_h} \frac{|b(\boldsymbol{v}_h, q_h)|}{\|\boldsymbol{v}_h\|_{1,h}} \qquad (q_h \in S_h). \tag{4.8}$$

Together with Lemmas 4.4 and 4.5, this proves that (4.1) has a unique solution [4]. Also, $\operatorname{div} \boldsymbol{u}_h^{\mathrm{div}} \in S_h$; testing the second equation with this divergence shows that it vanishes on every element.

4.3. Error equations. Assume $(\boldsymbol{u}, p) \in (\boldsymbol{H}^{k+1}(\Omega) \cap \boldsymbol{H}_0^1(\Omega)) \times (H^k(\Omega) \cap L_0^2(\Omega))$ solves (2.1). Set

$$\boldsymbol{u}_I^{\mathrm{div}}|_K = \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u}, \qquad \boldsymbol{u}_I^t|_F = \boldsymbol{\Pi}_{\partial K}^t \boldsymbol{u}, \qquad p_I = Q_h p.$$

For later use, (4.7), polynomial approximation, and inverse estimates imply

$$\begin{aligned} &\|\boldsymbol{u} - \boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u}\|_K + h_K |\boldsymbol{u} - \boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u}|_{1,K} \\ &\quad + h_K^{1/2} \|\boldsymbol{u} - \boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u}\|_{\partial K} \lesssim h_K^{k+1} |\boldsymbol{u}|_{k+1,K}. \end{aligned} \tag{4.9}$$

To see this, subtract a degree-k polynomial approximant and bound the remaining polynomial by its L^2 norm using inverse and trace estimates. The same argument applied to the strain gives

$$\begin{aligned} &\|\varepsilon(\boldsymbol{u}) - \varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u})\|_K \\ &\quad + h_K^{1/2} \|\varepsilon(\boldsymbol{u}) - \varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{I}_K^{\mathrm{div}} \boldsymbol{u})\|_{\partial K} \lesssim h_K^k |\boldsymbol{u}|_{k+1,K}. \end{aligned} \tag{4.10}$$

Since the normal interpolant trace and the tangential interpolant are face projections, (4.9) also gives

$$h_K^{-1/2} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I)\|_{\partial K} \lesssim h_K^k |\boldsymbol{u}|_{k+1,K}. \tag{4.11}$$

Set $e_h = \boldsymbol{u}_h - \boldsymbol{u}_I$ and $\xi_h = p_h - p_I$. The error equation is Write $\boldsymbol{E}_K = \varepsilon(\boldsymbol{u}) - \varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{u}_I^{\mathrm{div}})$ and $\boldsymbol{w}_K = \boldsymbol{v}_h^{\mathrm{div}} - \boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}}$. The error equation is

(4.12)

$$\begin{aligned} a_h(e_h, \boldsymbol{v}_h) + b(\boldsymbol{v}_h, \xi_h) = \mathcal{R}(\boldsymbol{v}_h) := \sum_{K \in \mathcal{K}_h} \big[&(\boldsymbol{E}_K, \varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}}))_K \\ &+ \langle \mathcal{J}_{\partial K}(\boldsymbol{v}_h), \boldsymbol{E}_K \boldsymbol{n} \rangle_{\partial K} + \langle Q_h p - p, \boldsymbol{w}_K \cdot \boldsymbol{n} \rangle_{\partial K} \\ &- \langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I), \varepsilon_K^w(\boldsymbol{v}_h) \boldsymbol{n} \rangle_{\partial K} \big]. \end{aligned}$$

for all $\boldsymbol{v}_h \in \boldsymbol{V}_h$. To derive it, test (2.1) with $\boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}}$, integrate by parts, and sum over K. The exact traction paired with $(\boldsymbol{v}_h^{\mathrm{div}} \cdot \boldsymbol{n})\boldsymbol{n} + \boldsymbol{v}_h^t$ cancels on interior faces and vanishes on the boundary. For the pressure terms, the volume projection preserves moments against $\nabla(Q_h p)$, and

$$(\operatorname{div} \boldsymbol{\Pi}_K^o \boldsymbol{v}_h^{\mathrm{div}}, p - Q_h p)_K = 0.$$

For the strain terms, use (3.14) twice and recall that $\varepsilon(\boldsymbol{\Pi}_K^o \boldsymbol{u}_I^{\mathrm{div}}) \in \mathbb{P}_{k-1}(K;\mathbb{S}) \subset \Sigma_h^{-1}(K)$. Its normal trace belongs to $V_k(F)$, so it pairs equally with the projected and full jumps. By (4.5), $\operatorname{div} \boldsymbol{u}_I^{\mathrm{div}} = 0$. Hence

$$b(e_h, q_h) = 0 \qquad \forall q_h \in S_h. \tag{4.13}$$

We estimate the four terms in (4.12) in order. The volume strain term is bounded by (4.10) and the norm of $\boldsymbol{v}_h$. The boundary strain term uses the same approximation estimate and the full-jump bound (3.15). For the pressure term, the local projection estimate is

$$h_K^{1/2} \|p - Q_h p\|_{\partial K} \lesssim h_K^k |p|_{k,K};$$

its other factor is the normal part of the jump, which the discrete norm controls directly. Finally, the polynomial trace inequality and the weak-strain upper bound give

$$\begin{aligned} &|\langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I), \varepsilon_K^w(\boldsymbol{v}_h)\boldsymbol{n}\rangle_{\partial K}| \\ &\quad \lesssim h_K^{-1/2} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I)\|_{\partial K} \|\boldsymbol{v}_h\|_{1,h,K}. \end{aligned}$$

Use (4.11), then sum the estimates over K. The Cauchy–Schwarz inequality yields

$$|\mathcal{R}(\boldsymbol{v}_h)| \lesssim h^k (\|\boldsymbol{u}\|_{k+1} + \|p\|_k) \|\boldsymbol{v}_h\|_{1,h}. \tag{4.14}$$

4.4. Error estimates. Taking $\boldsymbol{v}_h = e_h$ in (4.12) and $q_h = \xi_h$ in (4.13), coercivity gives

$$\|e_h\|_{1,h}^2 \lesssim a_h(e_h, e_h) = \mathcal{R}(e_h).$$

Therefore, (4.14) implies

$$\|e_h\|_{1,h} \lesssim h^k (\|\boldsymbol{u}\|_{k+1} + \|p\|_k).$$

For the pressure, (4.8) and (4.12) give

$$\begin{aligned} \|\xi_h\| &\lesssim \sup_{0 \neq \boldsymbol{v}_h \in \boldsymbol{V}_h} \frac{|\mathcal{R}(\boldsymbol{v}_h) - a_h(e_h, \boldsymbol{v}_h)|}{\|\boldsymbol{v}_h\|_{1,h}} \\ &\lesssim \|e_h\|_{1,h} + h^k (\|\boldsymbol{u}\|_{k+1} + \|p\|_k). \end{aligned}$$

Thus $\|\xi_h\|$ has the same bound.

In the velocity error norm, identify the smooth velocity with the pair $\{\boldsymbol{u}, \boldsymbol{\Pi}_{\partial K}^t \boldsymbol{u}\}$, and use the same norm formula with the L^2 volume projector. Polynomial approximation and (4.7) give $\|\boldsymbol{u} - \boldsymbol{u}_I\|_{1,h} \lesssim h^k \|\boldsymbol{u}\|_{k+1}$. The triangle inequality and $\|p - Q_h p\| \lesssim h^k \|p\|_k$ then yield

$$\|\boldsymbol{u} - \boldsymbol{u}_h\|_{1,h} + \|p - p_h\| \lesssim h^k (\|\boldsymbol{u}\|_{k+1} + \|p\|_k). \tag{4.15}$$

THEOREM 4.7 (Error estimate for the reconstructed viscous stress). *Under the assumptions of the preceding velocity–pressure error estimate, let* $\boldsymbol{\tau} = \varepsilon(\boldsymbol{u})$ *and* $\boldsymbol{\tau}_h = \varepsilon_h^w(\boldsymbol{u}_h)$. *Then*

$$\|\boldsymbol{\tau} - \boldsymbol{\tau}_h\|_{0,\Omega} \lesssim h^k \big(\|\boldsymbol{u}\|_{k+1,\Omega} + \|p\|_{k,\Omega}\big). \tag{4.16}$$

Proof. Let $\boldsymbol{u}_I$ be the velocity interpolant defined above, and write $\boldsymbol{G}_K = \varepsilon(\boldsymbol{\Pi}^o_K \boldsymbol{u}_I^{\mathrm{div}})$. Since $\mathbb{P}_{k-1}(K;\mathbb{S}) \subset \Sigma_h^{-1}(K)$, $\boldsymbol{G}_K$ belongs to the local stress space. The defining identity (3.14) therefore implies

$$\big(\varepsilon^w_K(\boldsymbol{u}_I) - \boldsymbol{G}_K, \boldsymbol{\eta}_h\big)_K = \big\langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I), \boldsymbol{\eta}_h \boldsymbol{n} \big\rangle_{\partial K} \qquad (\boldsymbol{\eta}_h \in \Sigma_h^{-1}(K)).$$

Taking $\boldsymbol{\eta}_h = \varepsilon^w_K(\boldsymbol{u}_I) - \boldsymbol{G}_K$ and using the polynomial inverse trace inequality on the local stress space gives

$$\|\varepsilon^w_K(\boldsymbol{u}_I) - \boldsymbol{G}_K\|_{0,K} \lesssim h_K^{-1/2} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\boldsymbol{u}_I)\|_{0,\partial K} \lesssim h_K^k |\boldsymbol{u}|_{k+1,K}, \tag{4.17}$$

where the last inequality is (4.11). Together with (4.10), this yields

$$\|\boldsymbol{\tau} - \varepsilon^w_h(\boldsymbol{u}_I)\|_{0,\Omega} \lesssim h^k \|\boldsymbol{u}\|_{k+1,\Omega}. \tag{4.18}$$

Finally, linearity of the weak strain, its upper stability bound, and the discrete velocity estimate proved above imply

$$\begin{aligned} \|\boldsymbol{\tau} - \boldsymbol{\tau}_h\|_{0,\Omega} &\le \|\boldsymbol{\tau} - \varepsilon^w_h(\boldsymbol{u}_I)\|_{0,\Omega} + \|\varepsilon^w_h(\boldsymbol{u}_I - \boldsymbol{u}_h)\|_{0,\Omega} \\ &\lesssim h^k \|\boldsymbol{u}\|_{k+1,\Omega} + \|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{1,h} \\ &\lesssim h^k \big(\|\boldsymbol{u}\|_{k+1,\Omega} + \|p\|_{k,\Omega}\big), \end{aligned}$$

as claimed. ∎

5. Numerical results. For $k = 1$, the velocity uses $\boldsymbol{V}_h$ and the pressure is piecewise constant on the primal mesh $\mathcal{K}_h$. The coefficient of the symmetric strain in (2.1) is one, which corresponds to $2\nu = 1$ in the usual viscosity convention. The weak strain (3.14) has the equivalent form

$$(\varepsilon^w_K(\boldsymbol{v}), \boldsymbol{\sigma})_K = -(\boldsymbol{v}^{\mathrm{div}}, \mathrm{div}_{w,K}\, \boldsymbol{\sigma})_K + \langle \boldsymbol{v}^{\mathrm{div}} \cdot \boldsymbol{n}, (\boldsymbol{\sigma}\boldsymbol{n}) \cdot \boldsymbol{n} \rangle_{\partial K} + \langle \boldsymbol{v}^t, \Pi_F(\boldsymbol{\sigma}\boldsymbol{n}) \rangle_{\partial K},$$

where $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$ with $k = 1$, and $\mathrm{div}_{w,K}\, \boldsymbol{\sigma} \in \mathbb{P}_1(K;\mathbb{R}^d)$ satisfies

$$(\mathrm{div}_{w,K}\, \boldsymbol{\sigma}, \boldsymbol{q})_K = -(\boldsymbol{\sigma}, \nabla \boldsymbol{q})_K + \langle \boldsymbol{\sigma}\boldsymbol{n}, \boldsymbol{q} \rangle_{\partial K} \qquad \forall \boldsymbol{q} \in \mathbb{P}_1(K;\mathbb{R}^d).$$

This definition includes the internal stress jumps through integration by parts on the cones. Its equivalence follows from the volume projection and the symmetry of $\boldsymbol{\sigma}$.

The reported `sincosdata` test is posed on the unit square. Tables 2–4 give the supplied results. The energy error is $\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{A_h} := a_h(\boldsymbol{u}_I - \boldsymbol{u}_h, \boldsymbol{u}_I - \boldsymbol{u}_h)^{1/2}$. The energy and total pressure errors approach first-order convergence on these meshes. Faster convergence in some other columns is an empirical observation; it is not proved by (4.15).

Remark 5.1 (Numerical data to confirm). The supplied files do not specify the exact velocity, pressure, force, boundary data, viscosity used in the computations, or the definitions of $\|\cdot\|_{0,h}$ and the reported maximum norms. The implemented strain space must also be checked against (3.1); the original numerical description used a different index. The table values are retained pending these checks. The rectangular meshes do not satisfy Assumption **??**: their face-tangential tensors span only diagonal symmetric matrices. Table 4 therefore reports a test outside the proved mesh class. The three mesh figure files were not supplied.

TABLE 2
Results on polygonal meshes obtained from a triangle-mesh dual.

DoFs	h	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{A_h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{0,h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{\infty}$	$\|p_I - p_h\|$	$\|p - p_h\|$	$\|p_I - p_h\|_{\max}$
314	1.43e-01	8.767e-01	2.588e-02	7.996e-02	2.071e-01	2.454e-01	5.002e-01
978	7.78e-02	4.012e-01	9.135e-03	2.645e-02	6.612e-02	1.007e-01	1.870e-01
3362	4.12e-02	1.884e-01	2.712e-03	7.111e-03	1.936e-02	4.463e-02	7.481e-02
12354	2.13e-02	9.119e-02	7.279e-04	1.817e-03	5.869e-03	2.145e-02	3.068e-02
47234	1.08e-02	4.490e-02	1.875e-04	4.573e-04	1.874e-03	1.061e-02	1.335e-02

TABLE 3
Results on triangular meshes.

DoFs	h	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{A_h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{0,h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{\infty}$	$\|p_I - p_h\|$	$\|p - p_h\|$	$\|p_I - p_h\|_{\max}$
232	2.50e-01	4.289e-01	1.463e-02	3.022e-02	5.357e-01	5.509e-01	8.008e-01
880	1.25e-01	1.949e-01	2.959e-03	6.035e-03	2.867e-01	2.940e-01	4.262e-01
3424	6.25e-02	9.410e-02	6.990e-04	1.479e-03	1.450e-01	1.486e-01	2.115e-01
13504	3.12e-02	4.659e-02	1.727e-04	4.138e-04	7.265e-02	7.446e-02	1.045e-01
53632	1.56e-02	2.323e-02	4.308e-05	1.098e-04	3.634e-02	3.725e-02	5.187e-02

TABLE 4
Results on rectangular meshes, outside Assumption **??**.

DoFs	h	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{A_h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{0,h}$	$\|\boldsymbol{u}_I - \boldsymbol{u}_h\|_{\infty}$	$\|p_I - p_h\|$	$\|p - p_h\|$	$\|p_I - p_h\|_{\max}$
152	2.50e-01	6.193e-01	4.377e-02	7.423e-02	1.557e-02	1.574e-01	2.227e-02
560	1.25e-01	2.438e-01	1.277e-02	1.582e-02	9.016e-03	8.021e-02	1.778e-02
2144	6.25e-02	1.083e-01	3.348e-03	4.464e-03	2.688e-03	4.011e-02	6.658e-03
8384	3.12e-02	5.211e-02	8.477e-04	1.148e-03	6.957e-04	2.004e-02	1.943e-03
33152	1.56e-02	2.579e-02	2.126e-04	2.889e-04	1.752e-04	1.002e-02	5.205e-04

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